

A Memory–Magic Exchange Law in Streaming Clifford+ T Compilation

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Abstract

A phase that reaches a fault-tolerant processor in several additive pieces can be remembered until the last piece arrives, or executed on arrival: the first option is paid for in classical bits carried across a round boundary, the second in magic states committed before the phase is known. We show that in the ancilla-free coordinatewise Clifford+ T model realized by per-rotation synthesis pipelines the two payments are convertible at an exchange rate α —committed T gates per bit of memory forgone—that does not depend on the problem size. Unconditionally $\alpha \geq 2$: it rests on a new equidistribution estimate for Clifford+ T words near rotation cosets, whose only external input is the Ramanujan bound of Lubotzky–Phillips–Sarnak and Parzanchevski–Sarnak, and which shows that a committed word of T -count τ carries at most $\tau/2 + O(\log \log(1/\varepsilon))$ bits about the share it commits. The bound is a statement about irreversible commitment rather than about the target: it holds unchanged when the aggregate phase turns out to be zero and the final unitary is the identity. An explicit fractional passthrough family attains $\alpha \rightarrow 3$ whenever the mean single-rotation synthesis cost on the grid is $(3 + o(1)) \log_2(1/\varepsilon)$, as Ross–Selinger synthesis gives for typical angles and as we measure at $\varepsilon = 10^{-10}$. The gap between 2 and 3 is the square-root remainder of the spectral estimate, and closing it is an arithmetic question: under an equidistribution conjecture whose exponent $\beta = 2$ is the codimension of a rotation coset in $SU(2)$ —with a linear additive term forced by the powers of infinite-order words, and which exhaustive enumeration to T -count 22 supports on Haar-random and on arithmetic cosets with exponents to three decimals and constants matching the Haar prediction to a few per cent—the rate is exactly $\alpha = 1 + \beta = 3$ and commitment is quantized: a committed word carries at most $O(\log \log(1/\varepsilon))$ bits about its coordinate unless it costs at least $2 \log_2(1/\varepsilon) - O(1)$ gates, so memory should be shed in whole rotations rather than low bits. All constants are explicit; at $\varepsilon = 10^{-10}$ the rigorous floor is 0.78 committed T gates per bit against 3.20 achieved by passthrough at the certified share accuracy. Side information held down-

stream enters through a conditional entropy: when the earlier shares are an injective function of a λ -bit seed independent of the aggregate, as in randomized compiling, the information a compiler without the seed is missing is λ rather than $m \log_2(1/\varepsilon)$.

1 Introduction

Two physical resources meet at the boundary between a fault-tolerant quantum processor and its classical control electronics. On the quantum side, every non-Clifford rotation consumes magic states, and the number of T gates is the dominant cost of a logical computation [1, 2]. On the classical side, an instruction that has been lowered into the pulse stream cannot be taken back, so the only quantity that a round of the computation can pass to the next one is the state it has decided, in advance, to carry [3, 4]. These are different resources with different exchange rates against everything else, and there is no a priori reason for them to be convertible into one another.

They become convertible when a phase arrives in pieces. A Trotter step, a randomized-compiling instance, a layerwise parametrized ansatz, or a phase-polynomial emitted by a scheduler presents the angle of a given rotation as a sum of shares delivered in different rounds. Whoever executes the stream then faces a choice at every share: hold it in classical memory until the sum is known and synthesize one rotation, or synthesize the share on arrival and consume the corresponding magic now. Our earlier work [5] established that the first option is charged in bits: any streaming process that is correct to projective error ε on all r -round dispersed streams of m commuting phases must carry, across the last inter-round cut, $\Omega(m \log(1/\varepsilon))$ bits of restart state or of committed output. The second option was there only estimated, through an empirical synthesis cost of about $3 \log_2(1/\varepsilon)$ T gates per rotation. The purpose of this paper is to make the second option a theorem and to determine the exchange rate.

The quantity traded against magic is not storage. A control processor has gigabytes of memory and the snapshot S below is measured in kilobytes—at $m = 10^4$ co-

ordinates and $\log_2 K = 33$ the entire aggregate is 41 kB, against the 10^6 magic states on the other side of the trade. What S measures is *deferred decision*: how much of the computation a schedule has refused to lower into the instruction stream until later rounds have been seen. The cost of not deferring is a property of the commitment and not of the target: under the canonical sharing the first $r - 1$ rounds are independent of the aggregate, so every lower bound below holds conditionally on any fixed aggregate, including the zero aggregate whose target unitary is the identity (Corollary 1)—a computation that cancels completely can still have consumed $\Theta(m \log(1/\varepsilon))$ magic states before the cancellation was knowable. Deferral is not free in three situations that occur in practice. In a real-time control stack the stream is append-only and the scheduler emits with bounded lookahead [3, 4], so a rotation that is not held must be committed; lookahead depth and snapshot size are different resources, and it is the second that is priced here. In randomized compiling and in any protocol where the shares are produced by different agents, the compiler does not hold the seed that generated the earlier shares; Theorem 6 shows that what it is then charged for is the entropy it is missing—for shares that are an injective function of a λ -bit seed independent of the aggregate, between $\lambda - m \log_2(Q/K)$ and λ , not $m \log_2 K$ (Corollary 4)—which is the precise boundary of the “randomize and lower late” rule of [5]. And when the angles are produced on the quantum side—qubitization, a QROM-loaded phase table, adaptive QSVT phases—no classical snapshot exists to be traded away at all; our proof, which applies Fano’s inequality to a classical stream, does not reach that case, and we state it as the natural target for this line of work (Sec. 11). The law is specific to the synthesis model: Sec. 10 records that the entry fee disappears in the phase-gradient model and, at small angles, in mixed-channel synthesis, and that the ancilla-assisted case is open.

Our object is the *committed magic* T_{pre} : the number of T gates executed before the last share of the stream has arrived. A process that holds every share in memory has $T_{\text{pre}} = 0$; a memoryless process must commit each share as it comes. The main results, stated for the ancilla-free coordinatewise Clifford+ T model of Sec. 2, are as follows.

- **Theorem 1** (unconditional). $T_{\text{pre}} \geq (r - 1)(m \log_2 K - S) - O(rm \log \log(1/\varepsilon))$, where S is the worst-input expected cut budget of Definition 1 and $K = \Theta(1/\varepsilon)$: one committed T gate per bit not held, per round.
- **Theorem 2** (equidistribution). The number of Clifford+ T words of T -count $\leq k$ within ε of any coset $G_1 R_z(\theta) G_2$ of a rotation subgroup is at most $C_1 2^k \varepsilon^2 + C_2(k+1)2^{k/2}$ with $C_1 \leq 408$ and $C_2 \leq 232$.
- **Theorem 3** (rate two). Consequently $T_{\text{pre}} \geq 2(r - 1)(m \log_2 K - S - mA_2)$ up to $O(\delta)$ corrections, where

$A_2 = \log_2 C_2 + 2 \log_2(2 \log_2 K) + O(1)$ bits per coordinate: every bit forgone costs at least two committed T gates, once the first A_2 bits of each coordinate are given away.

- **Theorem 4** (achievability). A fractional passthrough scheme attains $T_{\text{pre}} = (r - 1)(m - q) \bar{\tau}_Q(\varepsilon/r)$ at $S = q \log_2 Q$ for every q , where $\bar{\tau}_Q$ is the mean single-rotation synthesis cost on the grid, hence rate $\rightarrow 3$ whenever $\bar{\tau}_Q(\varepsilon/r) = (3 + o(1)) \log_2 K$, which Ross–Selinger synthesis gives for typical angles and which the grid samples of Sec. 9 support.
- **Conjecture 1 and Theorem 5** (rate $1 + \beta$, quantization). Under a stated equidistribution conjecture—whose exponent β is the codimension of a rotation coset in $\text{SU}(2)$, with a linear additive term that the powers of infinite-order words force (Proposition 1), and which the enumeration supports with Haar-exact constants—the rate is $1 + \beta = 3$, and a committed word carries at most $O(\log \log(1/\varepsilon))$ bits about its coordinate unless it costs at least $2 \log_2(1/\varepsilon) - O(1)$ gates: 7 bits at $\varepsilon = 10^{-10}$, against the 33 a coordinate carries.
- **Theorems 6 and 7** (side information). With side information Y available downstream, $m \log_2 K$ is replaced by the conditional entropy $H(X | \text{other rounds}, Y)$, with a one-shot version in terms of smooth entropies.

The two constants 2 and 3 are not the same kind of number. A single-qubit Clifford+ T word of T -count t is one of $36 \cdot 2^t$ unitaries (Matsumoto–Amano), so a T gate carries at most one bit of angle: that is the rate 1 of Theorem 1. What changes it is that the committed word is not free in the group but confined to a tube of radius ε around a rotation coset, a curve of codimension $\beta = 2$ in the three-dimensional group. If the words in that tube are distributed as volume predicts, a word of T -count τ can take one of $\approx 2^\tau \varepsilon^\beta$ values there, so carrying b bits costs $\tau \geq b + \beta L$: a marginal rate of one gate per bit on top of an entry fee of βL gates, which over the $\log_2 K \approx L$ bits of one coordinate averages to $1 + \beta = 3$. This is the same count as the Ross–Selinger cost $3L$ of a single rotation, seen from the counting rather than the synthesis side. What we can prove is weaker in a specific way: pointwise counting through Hecke operators controls the tube only up to the square root of that main term (Remark 2), leaving a profile $2^{\tau/2}$, under which carrying b bits costs $\tau \geq 2b$ —no entry fee, but a marginal rate $1 + \beta/2 = 2$. Rate 2 is therefore the natural threshold of the spectral method rather than a geometric constant, and closing the gap to $1 + \beta$ is an arithmetic problem, not a compilation one. Both statements are asymptotic in L ; Sec. 9 evaluates the additive terms at $\varepsilon = 10^{-10}$, where the unconditional floor is 0.78 and the conditional one 1.61 committed T gates per bit against 3.20 achieved.

What is new. Relative to [5], whose bit-level bound Theorem 1 reproves with the Matsumoto–Amano count in place of Kraft’s inequality, the new unconditional results are Theorem 2, a count of Clifford+ T words near a rotation coset with explicit constants, and Theorem 3, which turns it into the rate 2: the counting results of [10] bound covering exponents in the whole group, and Selinger’s [7, Thm. 30] bounds the worst-case cost of one angle; neither controls how many cheap words lie near a coset, which is what a streaming lower bound needs. Corollary 1 and Theorems 6–7 are new and elementary. Theorem 5 is conditional on Conjecture 1, whose exponent is fixed by geometry and whose constants we test but do not prove; Theorem 4 is an elementary construction whose rate depends on a synthesis hypothesis, Corollary 3. The rigorous content of the paper is therefore the pair (Theorems 3, 4): two gates per bit are necessary in this model and three are sufficient under the synthesis hypothesis, with every additive term explicit.

2 Setting

The model is the streaming compilation model of [5]. So that the present paper can be read on its own, Appendix A restates the four items we use—the charged one-pass process and its snapshots, the dispersed family, the closed form of the projective distance, and the block compiler—as Definitions 1 and 2, Lemma 7 and Proposition 2, with proofs. Nothing below depends on any other result of [5]; in particular Theorem 1 reproves its bit-level bound.

Family. Fix a modulus $Q \geq 3$, step $\Delta = 2\pi/Q$, alphabet size $K = Q - 1$, and m $\mathbb{F}_2$ -independent Z -characters $P_1, \dots, P_m$ on $n \geq m$ qubits; let V be a fixed public Clifford with $VZ_jV^\dagger = P_j$. The aggregate is $x \in [K]^m$, $[K] = \{0, \dots, K - 1\}$, and the target is

$$U_x = V \left(\bigotimes_{j \leq m} R_z(\Delta x_j) \otimes I \right) V^\dagger, \quad R_z(\theta) = e^{-i\theta Z/2}. \quad (1)$$

An r -round dispersed stream is a round-major sequence of shares $a^{(1)}, \dots, a^{(r)} \in \mathbb{Z}_Q^m$ with $\sum_t a^{(t)} \equiv x \pmod{Q}$. The *canonical sharing* $\mathcal{D}$ takes $a^{(1)}, \dots, a^{(r-1)}$ i.i.d. uniform on $\mathbb{Z}_Q^m$ and $a^{(r)} = x - \sum_{t < r} a^{(t)}$. The accuracy satisfies $0 < \varepsilon < \sin(\pi/2Q)$, so that distinct grid rotations are separated by $2\sin(\pi/2Q) > 2\varepsilon$ in the projective distance $d_{\text{proj}}(A, B) = \inf_\phi \|A - e^{i\phi} B\|$ and nearest-grid decoding is unambiguous. The *tuned* modulus $Q_\varepsilon := \lfloor \pi/(2\arcsin(2\varepsilon)) \rfloor$ is the largest for which the separation is at least 4ε ; it satisfies $\log_2 K_\varepsilon = L - \log_2(4/\pi) + o(1) = L - 0.35 + o(1)$, where throughout

$$L := \log_2(1/\varepsilon). \quad (2)$$

Output model. The *coordinatewise ancilla-free Clifford+ T model* (CW) is the one realized by per-rotation synthesis pipelines, in which each generator of the phase polynomial is handed to a single-qubit synthesizer and no ancilla or cross-coordinate cancellation is used [6, 5]: the write-once output is a list of records (j, g) with $j \in [m]$ and $g \in \{H, S, T\}$, the implemented unitary is $V(\bigotimes_j W_j)V^\dagger$ where W_j is the ordered product of the gates recorded on coordinate j , and the T -count is the number of T records. We say the output is *correct* if $d_{\text{proj}}(W_j, R_z(\Delta x_j)) \leq \varepsilon$ for every j . By Lemma 1, correctness of the whole unitary— $d_{\text{proj}}(V(\bigotimes_j W_j)V^\dagger, U_x) \leq \varepsilon$, the criterion of [5]—implies coordinatewise correctness, so every lower bound below also holds under the total criterion. The converse is not asserted: a construction with coordinate error ε need only have total error at most $m\varepsilon$, and for a total error budget ε_{tot} the share-by-share construction of Sec. 6 can use $\varepsilon_{\text{tot}}/(mr)$ per share.

Process and committed magic. The process is a compiler in the charged one-pass model of Definition 1: it reads the stream once, may use arbitrary RAM and a random tape whose state is part of every restart snapshot, and $S := \max_{\text{input}} \mathbb{E}_\rho \max_c B_{\text{cut}}(c)$ is the worst-input expected snapshot size. It is required to be correct with probability $\geq 1 - \delta$ on every x and every valid stream, $\delta < \frac{1}{2}$. Let T_t be the number of T records emitted while reading round t , $T_{\text{pre}} := \sum_{t < r} T_t$ the *committed magic*, and $T_{\text{out}} = T_{\text{pre}} + T_r$. Bars denote expectation over the tape, the canonical sharing and a uniform aggregate. We write $L_\delta(m, K) := (1 - \delta)m \log_2 K - h_2(\delta)$, $\ell_\delta := (1 - \delta) \log_2 K - h_2(\delta)$, and $\rho_m(\bar{T}) := m \log_2(e(1 + \bar{T}/m)) + 2 \log_2(1 + \bar{T}) + 1$.

3 Three lemmas

Lemma 1 (Localization). *For unitaries $A = \bigotimes_j A_j$, $B = \bigotimes_j B_j$ on the same tensor factors, $\max_j d_{\text{proj}}(A_j, B_j) \leq d_{\text{proj}}(A, B) \leq \sum_j d_{\text{proj}}(A_j, B_j)$. For 2×2 unitaries, $d_{\text{proj}}(A, B) = \sqrt{2 - |\text{tr}(B^\dagger A)|}$.*

Lemma 2 (Matsumoto–Amano count). *The number of single-qubit Clifford+ T unitaries modulo phase of minimal T -count exactly t is $n_1(0) = 24$ and $n_1(t) = 36 \cdot 2^t$ for $t \geq 1$; hence $\sum_{s \leq t} n_1(s) = 72 \cdot 2^t - 48$, and the number of m -tuples with total minimal T -count at most T is $N_m^{\leq}(T) \leq 36^m 2^T \binom{T+m}{m}$.*

Lemma 3 (Entropy versus T -count). *Let $M = (\bar{W}_1, \dots, \bar{W}_m)$ be a random m -tuple of single-qubit Clifford+ T unitaries and $T \geq \sum_j t_{\min}(\bar{W}_j)$ a random integer. Then $H(M) \leq \bar{T} + m \log_2 36 + \rho_m(\bar{T})$.*

Proofs are in Appendix B. Lemma 2 is the source of the exchange rate: a T gate carries at most $\log_2 2 = 1$ bit.

4 Committed magic: the unconditional bound

Theorem 1 (Committed magic). *For every process as in Sec. 2 and every round $t \in [r-1]$,*

$$S + \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t) \geq L_\delta(m, K). \quad (3)$$

Consequently

$$\begin{aligned} T_{\text{pre}}^- &\geq (r-1)[L_\delta(m, K) - S - m \log_2 36] \\ &\quad - (r-1)\rho_m\left(\frac{T_{\text{pre}}^-}{r-1}\right), \end{aligned} \quad (4)$$

and for the tuned family and $\delta = 0$, $T_{\text{pre}} \geq (r-1)(m \log_2 K_\varepsilon - S) - O(rm \log \log(1/\varepsilon))$.

Proof. Fix t . Let $A := a^{(t)}$, $\text{Past} := (a^{(s)})_{s < t}$, $\text{Fut} := (a^{(s)})_{s > t}$ (which includes round r), and $Z := (\text{Past}, \text{Fut}, \rho)$.

(i) $H(A \mid Z) = m \log_2 K$. Under $\mathcal{D}$ the joint law of $(a^{(1..r-1)}, x)$ is $Q^{-m(r-1)} P_X(x)$ with $a^{(r)}$ determined. Conditioning on $(\text{Past}, \text{Fut})$ fixes $c := \sum_{s \neq t} a^{(s)}$ and $x = A + c$, so the conditional law of A is proportional to $P_X(A + c)$, i.e. uniform on the translate $[K]^m - c$. The tape is independent of the stream.

(ii) *Message and Fano.* Let $M_t := (\bar{W}_1^{(t)}, \dots, \bar{W}_m^{(t)})$ be the tuple of unitaries (modulo phase) implemented by the gate segments emitted on the respective coordinates during round t , and σ_t the restart snapshot at the cut after round t . Given ρ the process is deterministic; the segments emitted before round t are a function of (Past, ρ) , and those emitted after the cut are a function of (σ_t, Fut) by the restart semantics of Definition 1(ii). Hence the final unitary on coordinate j , $U_j^{<t} \bar{W}_j^{(t)} U_j^{>t}$, is a function of (M_t, σ_t, Z) . On success it is within ε of $R_z(\Delta x_j)$, and nearest-grid decoding recovers x_j because neighbouring grid rotations are more than 2ε apart; thus $A = x - c$ is recovered with error probability $\leq \delta$. Fano's inequality gives $H(A \mid M_t, \sigma_t, Z) \leq h_2(\delta) + \delta \log_2(K^m - 1)$, i.e. $I(A; M_t, \sigma_t \mid Z) \geq L_\delta(m, K)$.

(iii) *Message size.* $I(A; M_t, \sigma_t \mid Z) \leq H(\sigma_t) + H(M_t)$. The snapshot serialization is prefix-free (Definition 1(iii)), so $H(\sigma_t) \leq \mathbb{E}|\sigma_t| \leq S$, the worst-input expectation dominating the average over inputs. The number T_t of T records emitted in round t is at least $\sum_j t_{\min}(\bar{W}_j^{(t)})$, so Lemma 3 gives $H(M_t) \leq \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t)$. This proves (3); summing over t and using concavity of ρ_m proves the rest. $\square$

Remark 1 (Multiple passes). For p forward passes, the alternating-simulation transcript—all $2p-1$ crossing snapshots together with the unitary tuple committed while scanning rounds $< r$ in any pass, cut between rounds $< r$ and round r as in [5]—determines the output, and the same argument gives $\Sigma_p + \bar{T}_A + m \log_2 36 +$

$\rho_m(\bar{T}_A) \geq L_\delta$ with $\Sigma_p \leq (2p-1)S$. The per-round splitting, and with it the factor $r-1$, is a one-pass phenomenon: a process that rereads round t after seeing round r need not commit anything during round t .

Corollary 1 (Commitment before cancellation is knowable). *Under the canonical sharing $\mathcal{D}$ the rounds $a^{(1)}, \dots, a^{(r-1)}$ are i.i.d. uniform and independent of the aggregate x , and T_{pre} is a function of $(a^{(1)}, \dots, a^{(r-1)}, \rho)$ alone. Hence for every fixed aggregate $x \in [K]^m$,*

$$\mathbb{E}[T_{\text{pre}} \mid X = x] = T_{\text{pre}}^-$$

and the lower bounds of Theorems 1, 3 and 5 hold after conditioning on $X = 0$, for which $U_x = I$. A process that is required to be correct on every valid stream, and is not promised in advance that the aggregate vanishes, commits the same expected magic whether or not the phases cancel.

Proof. Independence is item (i) of the proof of Theorem 1; T_{pre} counts records emitted while reading rounds $< r$, which by Definition 1(ii) depend only on those rounds and on ρ . $\square$

The corollary is a restatement of the sharing distribution, not an independent strengthening of the bound. A compiler that is told the promise $X = 0$ can emit nothing; the statement concerns the resources a universally correct process has already spent on a stream that turns out to cancel, and it separates committed cost from the intrinsic synthesis cost of the completed target, which for the identity is zero.

Theorem 1 is the bit-level theorem of [5] with Lemma 2 in place of Kraft's inequality; its content is that the counting rate $\log_2 2 = 1$ is the worst case. The rest of the paper is about the fact that the true rate is larger.

5 Words near rotation cosets

Arithmetic structure. Single-qubit Clifford+ T modulo phase is the full $\{(\sqrt{2})\}$ -arithmetic group Γ of the maximal order

$$\mathcal{O} = \mathcal{O}_K \oplus \mathcal{O}_K \frac{1+i}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+j}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+i+j+k}{2}, \quad \mathcal{O}_K = \mathbb{Z}[\sqrt{2}], \quad (5)$$

of the definite quaternion algebra $D = (-1, -1)_{\mathbb{Q}(\sqrt{2})}$ [9, 10]. D splits at the prime $P = (\sqrt{2})$; the completion $K_P = \mathbb{Q}_2(\sqrt{2})$ has residue field $\mathbb{F}_2$, and Γ acts on the 3-regular Bruhat–Tits tree X_P with vertex stabilizer $U = \mathcal{O}^\times / \mathcal{O}_K^\times \cong S_4$, the single-qubit Clifford group modulo phase, of order 24 [10, §4.1.3]. *T-count equals tree distance:* the sphere of radius t has $3 \cdot 2^{t-1}$ vertices, and $24 \cdot 3 \cdot 2^{t-1} = 36 \cdot 2^t$ is exactly Lemma 2. Let $\Lambda_k \subset \Gamma$ be the ball of T -count $\leq k$, $N_k = |\Lambda_k| = 72 \cdot 2^k - 48$, and for a left- U -invariant $h \in L^2(\text{PU}(2))$ let $T_j h(g) := \sum_{\gamma U: d(\gamma v_0, v_0)=j} h(\gamma^{-1}g)$ be the Hecke operator of level j .

Input (R). On $L_0^2(U \setminus \text{PU}(2))$ one has $\|T_j\| \leq (j+1)2^{j/2}$ [10, Eq. (3.10)]. This is the Ramanujan bound: by Jacquet–Langlands the automorphic representations of $D^\times$ that occur here correspond to holomorphic Hilbert modular forms over $\mathbb{Q}(\sqrt{2})$ of parallel weight 2, hence to representations in the discrete series at both infinite places, for which the Ramanujan conjecture is a theorem at every finite place [14], following Deligne’s bound in the elliptic modular case [12] and the local–global compatibility of [13]. The non-backtracking version [10, Eqs. (5.2)–(5.3)] gives the same bound directly for the Matsumoto–Amano syllables HT and SHT .

Theorem 2 (Tube bound). *For all $G_1, G_2 \in \text{SU}(2)$, all $\varepsilon \in (0, 1/8]$ and all $k \geq 0$, with $T_\varepsilon(C) := \{W : \exists \theta, d_{\text{proj}}(W, G_1 R_z(\theta) G_2) \leq \varepsilon\}$,*

$$\#(\Lambda_k \cap T_\varepsilon(C)) \leq C_1 2^k \varepsilon^2 + C_2 (k+1) 2^{k/2}, \quad (6)$$

with the absolute constants $C_1 = 72 \cdot 4\sqrt{2} \leq 408$ and $C_2 = 24(1 - 2^{-1/2})^{-1} 2^{3/2} \leq 232$.

The proof (Appendix C) reduces the tube to a spherical cap via the Hopf fibration whose fibre is the coset, majorizes the cap by a Poisson kernel of width 2ε —whose spherical-harmonic coefficients are r^ℓ with $1 - r = 2\varepsilon$, so that both constants can be read off—and bounds the count by the Hecke operator through a rank-one sup-norm estimate.

Remark 2 (The square-root barrier). Any majorant of a cap of radius ε on S^2 has spectral mass up to degree $\gtrsim 1/\varepsilon$, and the Hecke bound (R) is applied to each degree separately, so the error term $(k+1)2^{k/2}$ in (6) cannot be reduced by this method. It exceeds the main term $C_1 2^k \varepsilon^2$ whenever $k < 4L + 2\log_2(k+1) - 2\log_2(C_1/C_2)$, which for $k \geq 1$ includes the whole range $k \leq 4L$, and in particular the range $k \in [2L, 3L]$ in which synthesized rotations live. The true count in this range (Sec. 9) is the main term; proving that is the content of Conjecture 1.

Corollary 2 (Alphabet profile). *In the setting of Theorem 1, conditional on $(\sigma_t = s, Z = z)$ and on the correctness of coordinate j , the unitary $\overline{W}_j^{(t)}$ committed on coordinate j during round t lies in $T_\varepsilon(C_{s,z,j})$, the tube around the coset with*

$$G_1 = (U_j^{<t})^\dagger, \quad G_2 = (U_j^{>t})^\dagger,$$

where $U_j^{<t}$ is the segment emitted before round t and $U_j^{>t}$, the segment emitted after the cut, is a function of (s, Fut) ; the grid shift $R_z(\Delta c_j)$ is absorbed into θ , whose admissible values form the full grid because the grid is invariant under translation by Δc_j . Hence, with $n(\tau) := C_1 2^\tau \varepsilon^2 + C_2 (\tau+1) 2^{\tau/2}$,

$$\#\{\overline{W}_j^{(t)} : t_{\min} \leq \tau\} \leq n(\tau),$$

and $\log_2 n(\tau) \leq \tau/2 + \log_2(\tau+1) + 1 + \log_2 C_2$ for every $\tau \leq 4L$.

Theorem 3 (Rate two). *Under (R), put $\tau_* := \lceil 2\log_2 K \rceil$ and*

$$A_2 := 1 + \log_2 Z, \quad Z := \sum_{\tau \leq \tau_*} n(\tau) 2^{-\tau/2}, \quad (7)$$

so that $A_2 = \log_2 C_2 + 2\log_2(2\log_2 K) + O(1) = O(\log \log(1/\varepsilon))$. Then for every round t ,

$$\bar{T}_t \geq 2[(1-2\delta)m\log_2 K - 2mh_2(\delta) - S - mA_2], \quad (8)$$

hence $T_{\text{pre}}^- \geq 2(r-1)[(1-2\delta)m\log_2 K - 2mh_2(\delta) - S - mA_2]$.

The proof (Appendix D) shows that, given the snapshot and the other rounds, a committed word of T -count τ carries at most $\tau/2 + A_2$ bits about its coordinate, and that words of cost above τ_* cannot help because a coordinate carries at most $\log_2 K \leq \tau_*/2$ bits. Theorem 3 bounds a rate; it contains no statement about quantization, because the profile $(\tau+1)2^{\tau/2}$ is not $O(1)$ at $\tau \leq 2L$. Nor is A_2 negligible in practice: at $\varepsilon = 10^{-10}$ it is 20.0 bits out of the $\log_2 K = 32.9$ bits a coordinate carries, and the rate-two bound is the larger of the two unconditional bounds only where fewer than 3.7 bits per coordinate are retained, i.e. near the memoryless end of the frontier (Sec. 9).

6 An achievable tradeoff family

Theorem 4 (Fractional passthrough). *For every $q \in \{0, \dots, m\}$ and every single-rotation synthesizer of cost $\tau(\varepsilon)$ per rotation to projective error ε , there is a deterministic zero-error one-pass CW process with*

$$\begin{aligned} S &\leq \lceil q \log_2 Q \rceil + O(\log(mQr)), \\ T_t &\leq (m-q)\tau(\varepsilon/r) \quad (t < r), \\ T_{\text{out}} &\leq (m+(r-1)(m-q))\tau(\varepsilon/r). \end{aligned} \quad (9)$$

Proof. Keep running sums modulo Q of q chosen coordinates (the block compiler of Proposition 2 with $p=1$); emit every share of the remaining coordinates as it arrives, synthesized to ε/r (a zero share costs nothing); in round r emit the aggregate rotation of each stored coordinate and the last share of each committed one. The coordinatewise error is at most ε by the right inequality of Lemma 1 and unitary invariance of d_{proj} . $\square$

Corollary 3. *Write $\tau(u; \eta)$ for the cost of the synthesis word selected for the angle $2\pi u/Q$ at accuracy η and $\bar{\tau}_Q(\eta) := Q^{-1} \sum_{u \in \mathbb{Z}_Q} \tau(u; \eta)$. Under the canonical sharing every pre-final share is uniform on $\mathbb{Z}_Q$, so the process of Theorem 4 has the exact expected committed magic*

$$T_{\text{pre}}^-^{\text{ach}} = (r-1)(m-q)\bar{\tau}_Q(\varepsilon/r), \quad (10)$$

and along the family $T_{\text{pre}}^-^{\text{ach}}/[(r-1)(m\log_2 K - S)] \leq \bar{\tau}_Q(\varepsilon/r)/\log_2 K \cdot (1+o(1))$. If the mean single-rotation

cost on the tuned grid satisfies $\bar{\tau}_Q(\varepsilon/r) = (3 + o(1)) \log_2 K$ —which Ross–Selinger synthesis gives for a typical angle, $\tau = 3L + O(\log L)$ [6], and which Sec. 9 measures directly on the grid at $\varepsilon = 10^{-10}$ —the family has asymptotic slope 3. The worst case is $4L + O(1)$, optimal to an additive constant by the lower bound $4L - 9$ of [7, Thm. 30]. The slope is within a factor $\rightarrow 3$ of Theorem 1 and $\rightarrow 3/2$ of Theorem 3; under Conjecture 1 it matches Theorem 5 to leading order in L . The family is achievable, not proven optimal: nothing below excludes a process strictly beneath it.

The endpoint gap $\log_2 Q$ versus $\log_2 K$ is the difference between storing a share, which lives in $\mathbb{Z}_Q$, and the entropy of an aggregate, which lives in $[K]^m$ (Proposition 2).

Remark 3 (The first bits of a coordinate are cheap). Fractional passthrough is not optimal for the first few bits, and the reason is the exact alphabet of Lemma 5. Suppose $4 \mid Q$. On any coordinate and in any round a process may commit the Clifford $S^k = T^{2k}$ with $k := \text{round}(4u/Q) \in \mathbb{Z}_4$, where u is the share, at no T cost, and carry in the snapshot only the residual $u - kQ/4$, an integer with at most $Q/4 + 1$ values: two bits per coordinate are shed for no magic at all. This is the $\log_2 36$ of Theorem 1, of which $\log_2 24$ bits are the Cliffords. If $8 \mid Q$ a third bit follows for half a T gate: commit T^j , $j := \text{round}(8u/Q) \in \mathbb{Z}_8$, at cost $j \bmod 2$, i.e. $\frac{1}{2}$ on average over a uniform share, and carry $u - jQ/8$, which has at most $Q/8 + 1$ values. The tradeoff family therefore starts at $(2(r-1)m \text{ bits}, 0)$ and passes through $(3(r-1)m \text{ bits}, (r-1)m/2)$, and only then continues with the passthrough slope; the three bits are exactly the allowance $\log_2(c_0 + 1) = 3.17$ of the c_0 term, and by Lemma 5 the eight T -powers are the only words of cost below $L - \frac{1}{2}$ that can be used this way. The construction needs the divisibility: for other Q the residual $u - jQ/8$ is a distinct rational for every u , so nothing is saved, and rounding it is not free, because the tuned grid spacing $2\pi/Q_\varepsilon \approx 8\varepsilon$ is already of the order of the error budget. The tuned Q_ε is in general not a multiple of 8; the points of Fig. 2 that use this construction are taken on the admissible grid $Q = 8\lfloor Q_\varepsilon/8 \rfloor$, which changes $\log_2 K$ by less than 2^{-L} per coordinate. Under Conjecture 1 the low-cost alphabet is larger— $N_0 = O(L)$ words of cost $\leq \tau_0$, the powers of a cheap infinite-order word among them (Proposition 1)—and Theorem 5 bounds what it can buy by $\log_2 N_0 = O(\log L)$ bits per coordinate per round, 7 bits at $\varepsilon = 10^{-10}$. The rate statements $\alpha \geq 2$ and $\alpha \rightarrow 3$ describe the remaining $\log_2 K - O(\log L)$ bits of each coordinate, which is where all but $O(m \log L)$ of the tradeoff lies.

7 Rate three and the quantum of commitment

Conjecture 1 (H). *There are constants c_0, c_1, c such that for all $G_1, G_2 \in \text{SU}(2)$, all $\varepsilon \in (0, 1/8]$, all moduli $3 \leq Q \leq Q_\varepsilon$ (i.e. $\sin(\pi/2Q) \geq 2\varepsilon$, the separation of the tuned family) and all $\tau \geq 0$,*

$$\begin{aligned} \#\{W : t_{\min}(W) \leq \tau, \exists d \in \mathbb{Z}_Q : \\ d_{\text{proj}}(W, G_1 R_z(\frac{2\pi d}{Q}) G_2) \leq \varepsilon\} \\ \leq c_0 + c_1 \tau + c 2^\tau \varepsilon^\beta, \quad \beta = 2. \end{aligned} \quad (11)$$

The three terms have three different origins, and the constants are chosen for the domain stated. The exponent β is the codimension of a rotation coset in the three-dimensional group, and Haar measure predicts a cumulative coefficient $96\varepsilon Q/\pi \leq 24$ for $Q \leq Q_\varepsilon$ in (11)—twice the shell constant $c_{\text{grid}} \leq 12$ of Sec. 9, since $\sum_{s \leq \tau} 2^s \approx 2^{\tau+1}$; the restriction $Q \leq Q_\varepsilon$ is part of the statement because the accuracy condition alone admits grids up to twice as fine, on which the Haar coefficient doubles. A uniform constant must also absorb the finite-cost overshoot above onset that the identity, Clifford and axis-aligned frames show in Sec. 9 (shell ratios up to 1.75), and we take $c = 48$, twice the Haar value. Over every frame of Table 2, $\varepsilon \in \{2^{-3}, \dots, 2^{-8}\}$ and $\tau \leq 16$, the largest value of $(N - c_0 - c_1 \tau)/(2^\tau \varepsilon^2)$ is 43.5, on the axis-aligned frame at $L = 5$, $\tau = 11$; the identity coset at $\varepsilon = 1/8$, $\tau = 10$, where the cumulative count is 466, requires 27.4. Both lie below 48, and both lie above the Haar value 24, which is why the conjecture cannot carry it. The constant $c_0 \leq 8$ counts the words lying exactly on an arithmetic coset, which are the powers of T in the (G_1, G_2) frame (Lemma 4). The linear term is forced: a coset can contain, or lie within ε of, every power of an infinite-order word (Proposition 1), and the cheapest such word, HT , has $t_{\min}((HT)^j) = j$ because $(HT)^j$ is its own Matsumoto–Amano normal form, so $c_1 \geq 2$ from the powers j and $-j$; we take $c_1 = 2$. The words on or near a rotation coset thus consist of a volume-law population and a single sparse cluster, and the cluster is what a uniform statement must pay for. By Remark 2 the conjecture lies beyond (R): in the parametrization of Appendix B it asserts that among the $\xi = tt^\dagger \in \mathbb{Z}[\sqrt{2}]$ in a thin strip, those with $2^k - \xi$ also a relative norm have density $2^{-k+o(k)}$ —a shifted-norm sieve at the ramified prime $(1 - \omega)$.

Lemma 4 (Words on an arithmetic coset). *For $G_1, G_2 \in \Gamma$ the Clifford+ T unitaries lying exactly on $\{G_1 R_z(\theta) G_2\}$ are the eight words $G_1 T^j G_2$.*

Proof. $R_z(\theta) = G_1^{-1} W G_2^{-1} \in \Gamma$ is diagonal; in the parametrization of Lemma 5 this is the case $t = 0$, whence $u = \omega^a (1 - \omega)^{2k}$ up to a unit and the unitary is $\text{diag}(\omega^a, \omega^b)$ modulo phase. $\square$

Proposition 1 (The linear term cannot be removed). *No constants c_0, c satisfy (11) with $c_1 = 0$, even when $Q = Q_\varepsilon$ is tuned and even when $G_1, G_2 \in \Gamma$.*

Proof. Let $V := e^{-i\pi/4}HTHT$, of T -count 2, with rotation angle θ , $\cos(\theta/2) = (2 + \sqrt{2})/4$. Since $2\cos(\theta/2)$ is a root of $2y^2 - 4y + 1$ and not an algebraic integer, V has infinite order, and $I, V, \dots, V^N$ are distinct with $t_{\min}(V^j) \leq 2j$. Write $V = DR_z(\theta)D^\dagger$. Let $x := \theta/2\pi$ and let p/Q run over the convergents of x , $|x - p/Q| < Q^{-2}$; put $\varepsilon_Q := \frac{1}{2}\sin(\pi/2Q)$, so that $Q_{\varepsilon_Q} = Q$, and $d_j := jp \bmod Q$. Then $d_{\text{proj}}(V^j, DR_z(2\pi d_j/Q)D^\dagger) \leq \frac{j}{2}|\theta - 2\pi p/Q| < \pi N/Q^2$, which is below $\varepsilon_Q/2 \geq 1/(4Q)$ once $Q > 4\pi N$. For $G_1, G_2 \in \Gamma$: Γ is dense in $\text{PU}(2)$, so there is $G_1 \in \Gamma$ with $D = G_1 E$, $d_{\text{proj}}(E, I) \leq \varepsilon_Q/4$; then $d_{\text{proj}}(V^j, G_1 R_z(j\theta)G_1^{-1}) \leq 2d_{\text{proj}}(E, I) \leq \varepsilon_Q/2$, and with $G_2 := G_1^{-1}$ all $N+1$ powers are within ε_Q of a grid rotation on the arithmetic coset $G_1 R_z G_1^{-1}$. With $\tau = 2N$ the left side of (11) is $\geq N+1$ while, along the convergents, $c2^{2N}\varepsilon_Q^2 \rightarrow 0$; so $N+1 \leq c_0$ for every N , which is absurd. (At $p/Q = 1746/10009$, $N = 9$ and the frame D itself, a direct check gives ten words against $8 + 12 \cdot 2^{18}\varepsilon_Q^2 = 8.02$.) $\square$

Conjecture 1 is the quantitative form of an observation of Selinger [7, §9]: the elementary count $36 \cdot 2^\tau \geq c\varepsilon^{-3}$, which gives $3L - O(1)$ for a generic element of the three-dimensional group, “is not a priori clear” for z -rotations, since these form a one-dimensional submanifold; [7, Thm. 30] then proves the worst-case bound $4L - 9$ for a specific angle. What Conjecture 1 adds is a bound for *every* coset and *every* cost, with the constants, which is what a counting argument against a stream needs. It is also falsifiable in a useful direction. The ε -balls of distinct grid rotations are disjoint, so (11) bounds the number of grid rotations reachable at cost τ by $c_0 + c_1\tau + c2^\tau\varepsilon^2$; covering the $K_\varepsilon \approx \pi/4\varepsilon$ rotations of the tuned family then forces

$$\max_d \tau_{\min}(d) \geq 3L - \log_2(4c/\pi) - o(1) = 3L - 5.93 - o(1), \quad (12)$$

so Conjecture 1 cannot coexist with an ancilla-free synthesis of grid rotations below the Ross–Selinger exponent. The enumeration gives $\max_d \tau_{\min} = 10, 16, 18$ against $3L = 12, 15, 18$ at $L = 4, 5, 6$.

Theorem 5 (Rate three; quantization). *Assume Conjecture 1. Put $k := \log_2 K$, $b := \beta L - \log_2 c - 1$, $\tau_0 := \lfloor \beta L - \log_2 c \rfloor$, $N_0 := c_0 + c_1\tau_0 + 1$, $a_0 := 1 + \log_2 N_0$, $\kappa := 1 + b/k$ and $D_t := (1 - 2\delta)mk - 2mh_2(\delta) - S$. For every round $t < r$,*

$$\bar{T}_t \geq \kappa(D_t - ma_0) - 3m \log_2(1 + \bar{T}_t/m), \quad (13)$$

and, writing $\bar{A}_t^>$ for the expected number of coordinates that are correct and whose word committed in round t has minimal T -count greater than τ_0 ,

$$\bar{A}_t^> \geq \frac{D_t - ma_0}{k}. \quad (14)$$

With $\beta = 2$, $c = 48$, $c_1 = 2$, $c_0 = 8$ and $Q \leq Q_\varepsilon$, $\kappa \rightarrow 1 + \beta = 3$, so $\bar{T}_{\text{pre}} \geq 3(r-1)(m \log_2 K - S)(1 - o(1)) - O(rm \log L)$; a committed word of cost at most $\tau_0 = \lfloor 2L - 5.58 \rfloor$ carries at most $\log_2 N_0 = \log_2(2\tau_0 + 9) = O(\log L)$ bits about its coordinate—7.0 bits at $\varepsilon = 10^{-10}$, against the 32.9 a coordinate carries—and every bit beyond that allowance is carried by a word of cost at least $\tau_0 + 1$, i.e. $2L - \log_2 c - O(1)$ gates.

The proof is in Appendix D. The structural content is that memory cannot be shed a few bits at a time within a coordinate. The low-cost alphabet of cost $\leq \tau_0 \approx 2L$ holds at most $N_0 = O(L)$ words—the eight T -powers of Remark 3, which are real, and the powers of a cheap infinite-order word, which Proposition 1 shows must be allowed for: a process may align its prefix with the axis of HT and then encode an integer j by emitting $(HT)^j$, which is $\log_2 j$ bits for j gates—so it buys $O(\log L)$ bits, and a coordinate that carries more must commit a word of cost $\geq 2L - O(1)$; (14) counts how many coordinates per round must do so. The theorem does not say that every coordinate receiving a T gate has paid $2L$ gates—a process may mix cheap words with rare expensive ones—but the expected cost of every bit beyond the $O(\log L)$ allowance is κ gates, and the optimal way to pay it is to commit whole coordinates, which is what fractional passthrough does. The convergence $\kappa \rightarrow 3$ is slow: $\kappa = 2.82$ at $\varepsilon = 10^{-10}$ and 2.93 at $\varepsilon = 2^{-80}$, so the number that a resource estimate at a chemically relevant accuracy should use is κ , not 3.

Lemma 5 (Identity coset). *If $2\varepsilon^2 4^k < 1$, i.e. $k < L - \frac{1}{2}$, the only Clifford+ T unitaries with denominator exponent $\leq k$ within ε of some $R_z(\theta)$ are the eight powers of T .*

Lemma 5 (Appendix B) is an elementary instance of the fee: at the identity coset the first L gates buy nothing, unconditionally. The enumeration shows the true fee there is $2L + O(1)$: the least T -count ≥ 2 at which some word comes within ε of a grid rotation is 10, 10, 12, 17, 18 for $L = 4, \dots, 8$, i.e. between $2L$ and $2L + 3$.

8 Side information

Theorem 6 (Side information). *Let $X \sim P_X$ on $[K]^m$, let the sharing have any joint law with X , and let side information Y be available to the process from the start. Then for every round t ,*

$$\begin{aligned} S + \bar{T}_t + m \log_2 36 + \rho_m(\bar{T}_t) \\ \geq H(X \mid \text{Past}, \text{Fut}, Y) - h_2(\delta) - \delta \log_2(K^m - 1). \end{aligned} \quad (15)$$

Proof. Theorem 1 with $Z := (\text{Past}, \text{Fut}, Y, \rho)$; the tape is independent of (stream, Y) , and $a^{(t)} \leftrightarrow X$ is a bijection given the other rounds. $\square$

Corollary 4. (a) Uniform X , canonical sharing, trivial Y recovers Theorem 1. (b) If Y determines the earlier shares (the process holds the sharing or swirl seed), the right side is 0: no memory and no committed magic. This is the boundary of the design rule “randomize and lower late” of [5]. (c) Suppose the shares of rounds $1, \dots, r-1$ are a deterministic injective function of a λ -bit uniform seed V , independent of X and unknown to the process, and Y is trivial. Given the other rounds, $A^{(t)}$ is a function of V , so

$$H(X | \text{Past}, \text{Fut}) = H(A^{(t)} | \text{Past}, \text{Fut}) \leq H(V) = \lambda$$

and the bound of Theorem 6 degrades from $m \log_2 K$ to λ : this is the direction that matters in practice, since in randomized compiling λ is a handful of bits, not $m \log_2 K$. It degrades no further: for $r = 2$ and X uniform on $[K]^m$, $H(a^{(2)}) \leq m \log_2 Q$ and $H(a^{(2)} | A^{(1)}) = H(X) = m \log_2 K$, so $I(A^{(1)}; a^{(2)}) \leq m \log_2(Q/K)$ and

$$H(A^{(1)} | a^{(2)}) \geq \lambda - m \log_2(Q/K),$$

where $m \log_2(Q/K) = O(m/Q)$ is negligible for the tuned family. This is the precise form of [5, Rem. 9]: the lower bound is governed by the entropy the process is missing, not by the size of the aggregate. (d) For non-product sources Theorems 3 and 5 hold with $m \ell_\delta$ replaced by $\sum_j H(A_j | Z) - m(h_2(\delta) + \delta \log_2 K)$ and S by $S + \text{TC}(A | Z)$, since $\sum_j I(A_j; \sigma | Z) \leq I(A; \sigma | Z) + \text{TC}(A | Z)$.

Theorem 7 (One-shot). Let the process be deterministic with worst-case resources $S_{\max}$, $T_{\max}$, correct with probability $\geq 1 - \delta$ over $(X, \text{sharing}, Y)$, and $r = 2$. With the set-smoothed entropy $H_0^\delta(U | VY) := \min_{\Omega: \Pr[\Omega] \geq 1-\delta} \log_2 \max_{v,y} |\{u : (u,v,y) \in \Omega\}|$ of Renner–Wolf [15],

$$\begin{aligned} S_{\max} + T_{\max} + m \log_2 72 + m \log_2(e(1 + T_{\max}/m)) \\ \geq H_0^\delta(U | V, Y) \geq H_{\max}^\delta(U | V, Y). \end{aligned} \quad (16)$$

For i.i.d. coordinates, $H_{\max}^\delta(U^m | V^m Y^m) = mH(U | VY) + \sqrt{mV} \Phi^{-1}(1-\delta) + O(\log m)$ [16], V the conditional information variance. The Shannon form (Theorem 6) applies to expected resources and the one-shot form to worst-case resources; the two are not to be mixed.

9 Numerical evidence

We enumerated all single-qubit Clifford+ T unitaries of T -count ≤ 22 (3.0×10^8 words, of which 1.5×10^8 lie in the top shell $t = 22$) through the Matsumoto–Amano normal form, with the 24 right Cliffords verified by closure.

Conjecture 1 (Fig. 1). For each $t \leq 22$, $\varepsilon \in \{2^{-4}, \dots, 2^{-8}\}$ with tuned $Q_\varepsilon \in \{12, 25, 50, 100, 201\}$, and three coset pairs (identity and two Haar-random

Table 1: Grid counts on Haar coset pair 0; exact Haar prediction $c_{\text{grid}} 2^{t-2L}$, $c_{\text{grid}} = 48\varepsilon Q_\varepsilon / \pi$, in parentheses.

t	$\varepsilon = 2^{-4}$	2^{-5}	2^{-6}	2^{-7}	2^{-8}
12	187 (183)	48 (48)	11 (12)	2 (3.0)	0 (0.7)
15	1470 (1467)	372 (382)	100 (95)	23 (24)	4 (6.0)
19	23426 (23468)	6104 (6112)	1487 (1528)	338 (382)	77 (96)
22	187707 (187747)	48772 (48892)	12321 (12223)	3039 (3056)	786 (768)

(G_1, G_2)), we counted words within ε of some grid rotation $G_1 R_z(2\pi d/Q) G_2$ and words within ε of the whole coset. A least-squares fit over the 136 cells with count ≥ 30 gives

$$\log_2 N = 1.001 t - 2.007 L + \text{const}, \quad (17)$$

i.e. exponents (1, 2) to three decimals. The constants are not fitted. The grid constant is $c_{\text{grid}} = 36 \cdot \frac{1}{3} \cdot 8\varepsilon / (2\pi/Q_\varepsilon) = 48\varepsilon Q_\varepsilon / \pi$, which equals 12 only up to the rounding of Q_ε and is 11.46, 11.94, 11.94, 11.94, 12.00 for $\varepsilon = 2^{-4}, \dots, 2^{-8}$; the tube constant is 36 for every ε . At $t = 22$, pooling the two Haar pairs, $N/(2^t \varepsilon^2) = 11.47, 11.96, 11.99, 12.16, 12.14$ for the grid and 35.98, 35.98, 36.02, 36.20, 36.43 for the tube, agreeing column by column to 0.1, 0.2, 0.5, 1.8, 1.2 per cent and to 0.1, 0.1, 0.1, 0.6, 1.2 per cent respectively. Over the whole table, restricted to the 30 Haar-pair cells whose *expected* count exceeds 3000, N divided by its exact prediction lies in $[0.98, 1.04]$ for the grid and $[0.98, 1.01]$ for the tube; the cut must be on the expected and not the observed count, since selecting cells by observed count conditions on the upper Poisson tail. Using 12 in place of c_{grid} instead leaves the $\varepsilon = 2^{-4}$ column low by a systematic 4.5% at every t — at $t = 22$ that is 20 standard deviations in the single column of Table 1, and manifestly not Poisson noise. What the three extra T -counts buy is the small- ε end of the table, where the expected counts were previously of order one: at $t = 19$ the $\varepsilon = 2^{-8}$ cell held 77 words against a prediction of 96 (20% low, and Poisson-noisy), whereas at $t = 22$ it holds 786 against 768 (2% high). Under Haar measure the squared off-diagonal entry $|b|^2$ is uniform on $[0, 1]$, so the tube $\{|b|^2 \leq \varepsilon^2\}$ has measure ε^2 and $36 \cdot 2^t \varepsilon^2$ of the $36 \cdot 2^t$ words of T -count t are expected in it; the grid acceptance region $\{s + \varphi^2/4 \leq \varepsilon^2\}$ in the (transverse, angular) coordinates occupies one third of the tube per angular window of width 8ε , and the windows are spaced by $2\pi/Q_\varepsilon$, giving $c_{\text{grid}} 2^t \varepsilon^2$ with $c_{\text{grid}} = 48\varepsilon Q_\varepsilon / \pi$ as above. The words are therefore Haar-equidistributed near rotation cosets to within statistical error wherever the expected count is at least of order one. At the identity coset the count is exactly 0 for $2 \leq t \leq 9$ at $L = 4$, with onset at $t = 10 = 2L + 2$: that coset is *below* Haar at low cost, which is the fee.

Arithmetic cosets. The cosets that Corollary 2 hands to Theorems 3 and 5 are not Haar-random: G_1

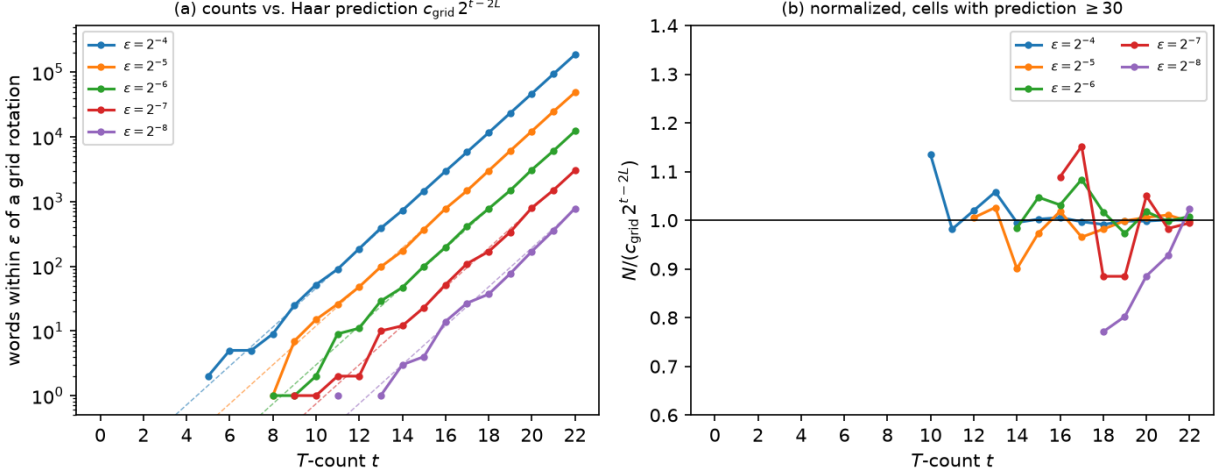


Figure 1: Number of Clifford+ T words of T -count exactly t within ε of a grid rotation on a Haar-random coset $G_1 R_z(\theta) G_2$. (a) Counts (markers) against the exact Haar prediction $c_{\text{grid}} 2^{t-2L}$ with $c_{\text{grid}} = 48\varepsilon Q_\varepsilon / \pi$ (dashed). (b) The same counts divided by that prediction, plotted where the prediction is at least 30; every ε has reached the Haar value by $t = 22$. Exhaustive enumeration to $t = 22$ (3.0×10^8 words, 1.5×10^8 in the top shell).

and G_2 are themselves Clifford+ T words, and Conjecture 1 is asserted uniformly in (G_1, G_2) , such frames included. We therefore repeated the whole count on four arithmetic pairs—random Matsumoto–Amano words of T -count 5, 10 and 15, and a pair of Cliffords (T -count 0)—with the three pairs of the previous paragraph recomputed as a regression (345 cells, all identical). The exponents do not move: a fit over the 180 arithmetic cells with count ≥ 30 gives $\log_2 N = 1.006t - 2.003L$ for the grid count and $1.002t - 2.009L$ for the tube, against $1.001t - 2.000L$ and $0.997t - 2.003L$ on the Haar pairs. Nor does the constant, for pairs of T -count 10 and 15: over all cells with expected count ≥ 100 the ratio $N/(c_{\text{grid}} 2^{t-2L})$ lies in $[0.87, 1.07]$, which is the spread of the Haar pairs themselves ($[0.87, 1.08]$). Cosets close to the identity behave differently, and exactly as they must. A Clifford pair reproduces the identity-coset counts *cell by cell*: for G_1, G_2 Clifford, $W \mapsto G_1^\dagger W G_2^\dagger$ is a T -count-preserving bijection of Λ_k carrying the grid onto itself, so Clifford cosets are isometric copies of the identity coset and inherit its fee. There the ratio ranges over $[0.34, 1.75]$ —suppressed below the onset $t \approx 2L$, overshooting for a few T -counts above it—and the T -count-5 pair interpolates, $[0.73, 1.24]$. All 70 cells at $t = 22$ lie in $[0.86, 1.04]$. Uniformity in (G_1, G_2) thus costs about a factor of two in the constant c ; since κ depends on c only through $\log_2 c$, Conjecture 1 carries $c = 48$ rather than the cumulative Haar value 24; the difference is $\kappa = 2.82$ against 2.85 at $\varepsilon = 10^{-10}$. The additive terms are a separate matter: an arithmetic coset contains the exact word $G_1 G_2$ at $\theta = 0$, so it has grid hits at T -count $\leq t_1 + t_2$ no matter how small ε is, and a frame whose axis is aligned with that of an infinite-order word carries the powers of that word

(Proposition 1). Table 2 counts, on each frame and each ε , the words of cost $\leq \tau_0 = \lfloor 2L - \log_2 24 \rfloor$ within ε of a grid rotation, against the $c_0 + c_1 \tau_0$ that Conjecture 1 allows. On the identity and Clifford frames the count is the number of T -powers that fall within ε of the grid; on the arithmetic and Haar-random frames it is at most three. The last row is a frame built for Proposition 1: $G_1 \in \Gamma$ of T -count 16 is the word of cost ≤ 16 whose image $G_1 \hat{z}$ of the z -axis is closest to the rotation axis of HT (angle 1.0×10^{-2}), and $G_2 = G_1^{-1}$, so that the powers $(HT)^{\pm j}$ lie within about 10^{-2} of the coset. Where that alignment is finer than ε ($L \leq 6$) the cluster is visible—seven words at $L = 4$, i.e. all of $(HT)^{\pm j}$ with $j \leq \tau_0 = 3$, and seven of the fifteen candidates at $L = 6$, about the fraction of the coset that the grid balls cover at the tuned modulus; where it is coarser ($L \geq 7$) the frame behaves like an arithmetic one. Every entry is below $c_0 + c_1 \tau_0$, and the linear term is a worst case over frames rather than the population of a generic one. The shell counts of the previous paragraphs come from `scripts/enum_arith.py` and `scripts/arith_analysis.py`; Table 2, the exact-optimal costs $\tau_{\min}(d)$ below, and Fig. 2 from `lowcost_counts.py`, `thm_numbers.py` and `make_fig2.py`.

Exact optimal synthesis and the frontier (Fig. 2). The same enumeration yields, for $L \leq 6$, the minimal T -count $\tau_{\min}(d)$ of a word within ε of every grid rotation $R_z(2\pi d/Q)$; this is the exact optimum, independent of factoring. At $L = 5$ ($Q = 25$) the mean synthesis cost over the 24 nonzero grid angles is 11.83 and the largest is 16 ($3L = 15$); at $L = 6$ ($Q = 50$) they are 15.59 and 18 ($3L = 18$). Fractional passthrough with canon-

Table 2: Words of cost $\leq \tau_0 = \lfloor 2L - \log_2 24 \rfloor$ within ε of a grid rotation of the tuned family, on the frames indicated, against the allowance $c_0 + c_1\tau_0$ of Conjecture 1 with $(c_0, c_1) = (8, 2)$. On the axis-aligned frame the powers $(HT)^{\pm j}$, $j \leq \tau_0$, lie within 10^{-2} of the coset, which is below ε at $L \leq 6$ and above it at $L \geq 7$. Computed by `scripts/lowcost_counts.py`.

L	4	5	6	7	8
τ_0	3	5	7	9	11
identity	4	3	2	4	3
Clifford pair	4	3	2	4	3
arithmetic, T -count 5	0	0	0	0	3
arithmetic, T -count 10	1	1	0	1	1
arithmetic, T -count 15	3	0	1	1	0
Haar-random pair A	0	0	0	0	1
Haar-random pair B	1	1	1	0	0
axis-aligned (Prop. 1)	7	3	7	3	3
$c_0 + c_1\tau_0$	14	18	22	26	30

ical sharing ($m = 64$, $r = 2$) then has $S = q \log_2 Q$ and $T_{\text{pre}} = (m - q)\mathbb{E}_u \tau_{\min}(u)$, in which u is uniform on all of $\mathbb{Z}_Q$ and the zero share is free, so the mean that sets the slope is $\mathbb{E}_u \tau_{\min} = 11.36$ and 15.28 — the $d \neq 0$ means above, times $(Q - 1)/Q$ — and the achieved slopes are $\mathbb{E}\tau/\log_2 K = 2.48$ ($L = 5$) and 2.72 ($L = 6$), approaching 3 from below as $(3L - 2.5)/(L - 0.35)$. A partial-commitment variant that commits the low ℓ bits of a share as a rotation and stores the rest—cost $\mathbb{E}_u \tau_{\min}(u \bmod 2^\ell)$ per coordinate for ℓ bits shed—costs 4.8, 3.2, 2.9, 2.7 T gates per bit shed for $\ell = 1, \dots, 4$ at $L = 5$, and 8.5, 6.1, 4.6, 3.7, 3.0 for $\ell = 1, \dots, 5$ at $L = 6$: strictly worse than passthrough and worsening as ℓ decreases, which is the quantum of commitment seen at exact optimality. At $L \leq 6$ the additive terms swamp the main terms: at $L = 6$ they are $m \log_2 36 = 5.17m$ for Theorem 1 and $m A_2 = 15.4m$ for Theorem 3, against $m \log_2 K = 5.6m$, so both bounds are vacuous, and so is Theorem 5, whose additive term $\kappa(1 + \log_2 N_0)$ already exceeds $\kappa \log_2 K$ at $L = 6$. Fig. 2(a) therefore tests slopes, not intercepts.

Chemistry scale (Fig. 2(b)). Exhaustive enumeration stops at $L \leq 6$, so at a chemically relevant accuracy we replace $\tau_{\min}$ by Ross–Selinger synthesis (`gridsynth`, whose ε is the up-to-phase operator norm and so is exactly d_{proj}). At $\varepsilon = 10^{-10}$ the tuned modulus is $Q_\varepsilon = 7853981633$ and $\log_2 K_\varepsilon = 32.871$. Under canonical sharing the share u is uniform on $\mathbb{Z}_Q$ and the frontier depends on it only through $\mathbb{E}_u \tau(u)$, so we sample rather than synthesize all Q_ε rotations: 20 000 uniform u give $\mathbb{E}_u \tau = 102.32 \pm 0.02$ (s.d. 2.16, range 88–113) against $3L = 99.66$, hence $\mathbb{E}\tau/\log_2 K = 3.113 \pm 0.001$; this is the single-rotation cost at accuracy ε , a calibration. The certified full-stream point of Theorem 4 with $r = 2$ synthesizes each committed share to $\varepsilon/2 = 5 \times 10^{-11}$; the same 20 000 angles synthesized at that accuracy give

$\mathbb{E}_u \tau = 105.34$, i.e. 3.205 per bit, three gates more per share as the $3 \log_2 r$ of Ross–Selinger scaling predicts, and it is this point that Table 3, Table 4 and Fig. 2(b) use. Both numbers exceed 3 slightly, as expected — $\tau_{\text{RS}} = 3L + O(\log L)$, and Corollary 3 asserts only the limit. With $m = 10^4$ and $r = 2$ this is $T_{\text{pre}} = 1.053 \times 10^6$ committed T gates at $S = 0$; both axes are linear in m , so $m = 10^5$ is the same line scaled tenfold. Partial commitment of the low ℓ bits, with the committed piece synthesized at ε , costs 61, 46, 35, 28, 22, 19, 16, 14 T gates per bit shed for $\ell = 1, \dots, 8$ (at $\varepsilon/2$ each entry rises by about three per share): the quantum of commitment persists at scale, at between 4.4 and 20 times the passthrough slope, because in this model a rotation by any nonzero grid angle, however small, costs a full synthesis — the small angles $2\pi k/Q$ with $0 < k < 2^\ell$ here average 110 to 122 T gates, slightly *above* the generic 102. The one exception is the exact commitment of Remark 3, which on the admissible grid $Q = 8\lfloor Q_\varepsilon/8 \rfloor$ sheds two bits per coordinate for no magic and a third for $\frac{1}{2}$ a T gate; those points, and not the $\ell = 1$ cross, are the true left end of the family. That the fee is a property of the deterministic-unitary model and not of fault-tolerant synthesis in general is the subject of Sec. 10.

The bounds at chemistry scale (Table 3). With C_1, C_2 explicit, the lower bounds can be evaluated rather than compared as rates. At $\varepsilon = 10^{-10}$ and $S = 0$, Theorem 1 gives $\bar{T}_t \geq 21.75m$, Theorem 3 gives $25.72m$ (with $A_2 = 20.0$ bits of the 32.9 available), Theorem 5 gives $52.9m$, and passthrough achieves $105.34m$ at the certified share accuracy: the conditional bound is within a factor 2 of achievability, the unconditional ones within a factor 4.1 (Theorem 3) and 4.8 (Theorem 1). The rate-two bound pays its additive term twice, so it exceeds the rate-one bound only for $S < 3.7m$ bits—11% of the range of S , at the memoryless end—and only because ε is small enough to begin with: at $S = 0$ the two agree at $L = 28.7$, and below that accuracy the rate-one bound is the stronger of the two. Fig. 2(b) plots these bounds themselves, not their leading-order slopes: because the bounds of Theorems 1 and 5 contain a logarithm of $\bar{T}_t$, their finite-precision derivatives $-\partial \bar{T}_t / \partial S$ at $S = 0$ are 0.94 and 2.61 rather than the leading coefficients 1 and $\kappa = 2.82$. All four numbers are per coordinate and per round, are computed by `scripts/thm_numbers.py`, and scale linearly in m .

10 Model dependence

Table 4 summarizes what changes with the synthesis model. Three entries deserve comment. In the phase-gradient model the exchange is bit-granular: adding the high bits of a share into the top bits of the gradient register is an exact operation whose cost is linear in the

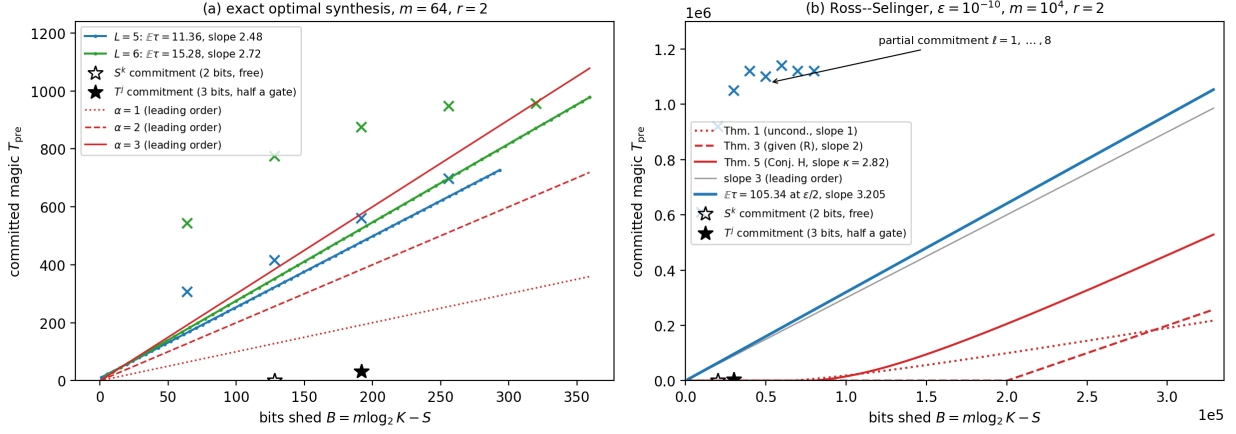


Figure 2: The fractional-passthrough tradeoff family ($r = 2$) against the lower bounds. (a) Exact minimal T -counts from the enumeration, $m = 64$, $L = 5, 6$ —an optimum no synthesis algorithm can beat—against the leading-order slopes $\alpha = 1, 2, 3$; at $L \leq 6$ the additive terms swamp the main terms, all three bounds being vacuous, so only slopes can be compared there. (b) Ross-Selinger synthesis at $\epsilon = 10^{-10}$, $m = 10^4$, from 20 000 sampled grid rotations, against those three bounds as stated, intercepts included (Table 3; the rate-three line uses the constants of Theorem 5 with $c = 48$, intercept $52.9m$); the achievable line is the $\epsilon/2$ point $\mathbb{E}\tau = 105.34$; the grey line is the leading-order slope 3, for comparison. The rate-two bound (dashed) overtakes the rate-one bound (dotted) only near the right-hand end, $S < 3.7m$ bits. Crosses: partial commitment of the low ℓ bits of each share. Crosses use the committed piece synthesized at ϵ . Star: the exact commitment of Remark 3 on the grid $8\lfloor Q_\epsilon/8 \rfloor$, three bits per coordinate for half a T gate (the first two of them free).

Table 3: The bounds at $\epsilon = 10^{-10}$ ($L = 33.22$, $Q_\epsilon = 7853981633$, $\log_2 K = 32.871$), per coordinate per round, at $S = 0$. “Coeff.” is the leading coefficient of $-S$ in the bound (for the bounds of Theorems 1 and 5 the finite-precision derivative $-\partial \bar{T}_t / \partial S$ at $S = 0$ is 0.94 and 2.61, because of their logarithmic terms); “per bit” is $\bar{T}_t / (m \log_2 K)$ at $S = 0$. The achievable row is the fractional-passthrough family with each committed share synthesized at $\epsilon/2$, the accuracy Theorem 4 requires for $r = 2$; the single-rotation calibration at ϵ is 102.32.

	$\bar{T}_t/m$	per bit	coeff.
Thm. 1 (unconditional)	21.75	0.66	1
Thm. 3 (given (R))	25.72	0.78	2
Thm. 5 (under Conj. 1, $c = 48$)	52.9	1.61	$\kappa = 2.82$
achievable (Thm. 4, gridsynth at $\epsilon/2$)	105.34	3.20	3.205

number of bits, so the fee of the CW model disappears; the catalytic gradient state is, in this sense, a device that has paid the fee once for all rotations. In the mixed-channel models now used in resource estimation the fee also disappears, but only at small angles and for a different reason: a rotation by θ may be realized as a probabilistic or quasi-probabilistic mixture of Clifford+ T channels in which the identity, at zero cost, carries most of the weight, giving an average cost $\tilde{O}(\theta^2/\delta)$ instead of $O(\log(1/\delta))$ [21]. The low bits of a share are exactly the small angles, so a mixing implementation can shed them cheaply and the quantization of Theorem 5 does not survive there. Our counting argument does not apply to channels—it counts deterministic words—and we make

no claim in that model; what the CW statements do say is that the effect is a property of mixing and not of magic. In the fully general n -qubit model even the premise fails: a T gate on n qubits can be any of $2 \cdot 4^n$ Pauli rotations, and with ancillas a tensor product of many single-qubit rotations can be synthesized for far less than the sum of their costs, the optimal T -count for an arbitrary diagonal n -qubit unitary being $\Theta(\sqrt{2^n \log(1/\epsilon)} + \log(1/\epsilon))$ [22]. Whether an $\Omega(m \log(1/\epsilon))$ committed-magic bound survives ancilla-assisted synthesis is the main question we leave open; the closest available tools are the stabilizer-nullity lower bounds of [20] and their adaptive extensions in [22]. Magic monotones do not settle it: they bound the magic of the final unitary, which is $\Theta(m)$, not the magic committed before the information arrived.

11 Discussion

In the model where rotations are synthesized coordinate by coordinate without ancillas, a bit of aggregate phase information that a schedule does not carry across a round boundary costs at least two committed magic states, and three suffice along the passthrough family whenever the grid-mean synthesis cost is $(3 + o(1))L$ —asymptotically in L , and after the first $O(\log L)$ bits of each coordinate. The two constants are not the same kind of number. Three is a volume count: a rotation coset has codimension $\beta = 2$ in $SU(2)$, so a word pays βL gates to enter the tube around it and one gate per bit thereafter, and $1 + \beta$ is also the Ross-Selinger cost of a single rotation. Two is

Table 4: Exchange rate α (committed T gates per bit of memory forgone) in several synthesis models.

Model		Rigorous lower bound	Achievable	Granularity
CW Clifford+ T	ancilla-free	1 (Thm. 1); 2 (Thm. 3, given (R)); $1 + \beta \rightarrow 3$ under Conj. 1 (Thm. 5)	$\tau_{\text{RS}}/\log_2 K \rightarrow 3$ (Thm. 4); exact 2.48 ($L = 5$), 2.72 ($L = 6$); 3.205 by gridsynth at the certified share accuracy $\varepsilon/2$, $\varepsilon = 10^{-10}$	one rotation (fee $2L - O(1)$) under Conj. 1, past the first $O(\log L)$ bits of a coordinate (Rem. 3, Prop. 1)
CW RUS/fallback/mixing (one ancilla, measurement)	+	not covered: the count concerns deterministic words, not channels	$\tau/\log_2 K$, $\tau \approx 1.15L$ [17], smaller with mixed fallback [18] and with (quasi-)probabilistic mixtures [21]	<i>no fee at small angles</i> : a mixture may put most of its weight on the identity, at average cost $\tilde{O}(\theta^2/\delta)$ [21], so the low bits of a share are nearly free
Phase gradient $Q = 2^b$	+ adder,	none for general prefixes	≈ 4 [19]: commit the high $b - \ell$ bits of a share into the top gradient bits, remember the low ℓ bits, one b -bit addition in the suffix	one bit, no fee: the catalytic gradient state pays the fee once
General Clifford+ T	n -qubit	$(m \log_2 K - S - \log_2 \mathcal{C}_n - O(n))/(2n+1)$ from $\# \leq \mathcal{C}_n (2 \cdot 4^n)^t$; degenerate for $n \sim m$	—	open
$Q \mid 8$		vacuous: $\log_2 K \leq \log_2 7 < \log_2 36$, so Theorem 1 asserts nothing	≤ 1 T gate per share: every grid rotation is a power of T , exactly	one bit, no fee. This case is <i>not</i> excluded by $\varepsilon < \sin(\pi/2Q)$, which at $Q = 8$ permits every $\varepsilon \leq 1/8$; it is the c_0 of Conj. 1 filling the whole grid

what the spectral method yields: the Ramanujan bound for the octahedral golden gates controls the tube only to the square root of its main term, and a profile $2^{7/2}$ is worth $1 + \beta/2$ gates per bit. The gap between them is the square-root remainder of the spectral estimate; whether the true rate is 3 is not decided here, and Conjecture 1—which the enumeration supports on Haar-random cosets, on arithmetic cosets, and, with a larger constant, on the Clifford cosets that are the worst case—is the statement that the arithmetic words do not know the difference.

Three consequences are worth stating plainly. First, the corner $S = 0$ is where the law binds without anyone choosing to be memoryless. When the rotation angles are produced on the quantum side—qubitization, a QROM-loaded phase table, adaptive QSVT phases—no classical snapshot of them exists to be traded away. If a reduction of that setting to the classical-input model existed, Theorem 3 would bind there with nothing subtracted for memory; none is given here, and the theorem is not to be applied to it: Fano’s inequality is applied to a classical stream, whereas with quantum coefficients the object crossing the cut is a state and not a message. That corner is the natural target for this line of work rather than a consequence of it. Extending the model to quantum coefficients is left to future work. Second, memory cannot be traded for magic in small units inside a coordinate: past the $O(\log L)$ bits that the eight T -powers and the powers

of a cheap infinite-order word can buy (Remark 3, Proposition 1), the trade is quantized in whole rotations, and a scheduler that must shed memory should shed whole coordinates rather than low bits—the enumeration puts the penalty for shedding low bits at between 4.4 and 20. Third, side information enters at the level of a conditional entropy: a process that holds the seed of the randomization faces no information obstruction, and one that misses an injective λ -bit seed independent of the aggregate is charged for between $\lambda - m \log_2(Q/K)$ and λ bits, not $m \log_2 K$ (Corollary 4(c)), which is the regime of randomized compiling, where λ is a handful of bits.

Two unconditional statements bound the arithmetic question from the other side: Selinger’s worst-case T -count $4L - 9$ for a specific z -rotation [7, Thm. 30], and the covering exponents of Parzanchevski and Sarnak, who show that some single-qubit unitaries need T -count $4L$ while all are reachable at $6L$ [10]. The committed-magic frontier is a statement about the typical rather than the worst case, and it sits between the rigorous and the conjectural covering exponents in the same way. Closing the gap between $1 + \beta/2$ and $1 + \beta$ is a problem about lattice points near a curve below the square-root barrier, and we leave it open.

Acknowledgments

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Data and code availability

The exhaustive enumeration of Clifford+ T words to T -count 22, the coset counts of Table 1 and Fig. 1, the exact-optimal and gridsynth data of Fig. 2, and the scripts `enum.arith.py`, `arith.analysis.py`, `lowcost_counts.py`, `thm_numbers.py` and `make_fig2.py` that produce every number and every figure quoted in Secs. 9 and 10 are available at <https://github.com/XXXX/memory-magic>.

A The streaming model

This appendix states the four items of [5] that Secs. 2–8 use, in the form in which they are used, with proofs, so that the present paper can be refereed on its own. The definitions are those of [5, Def. 9, 13, 20, 21] and the two statements are [5, Lem. 1, Prop. 10]; no other result of [5] is used anywhere above.

Definition 1 (Charged one-pass process). A *charged one-pass process* on the stream $a^{(1)}, \dots, a^{(r)}$ is an algorithm with a random tape ρ and a write-once output tape such that

- (i) it reads the stream once, in order, and may keep unbounded internal state while reading a round;
- (ii) at each *cut* c between consecutive rounds it emits a *snapshot* σ_c , a finite bit string, with the *restart* property: everything the process writes after c is a function of σ_c and of the part of the stream after c . The tape ρ is part of σ_c , so a process that wishes to reuse its randomness pays for it;
- (iii) snapshots are serialized in a prefix-free code, whence $H(\sigma_c) \leq \mathbb{E}|\sigma_c|$;
- (iv) its *charge* is $S := \max_{\text{input}} \mathbb{E}_\rho \max_c B_{\text{cut}}(c)$, where $B_{\text{cut}}(c) := |\sigma_c|$.

For $p > 1$ passes the process may re-read the stream p times, and the object playing the role of σ_c is the transcript of all $2p - 1$ crossings of the cut.

Condition (iii) is free: prefixing a snapshot with its own length in Elias- γ makes any serialization prefix-free at a cost of $2\log_2(|\sigma_c| + 1) + 1$ bits, which is dominated by the $O(\log(mQr))$ terms of every construction below. Condition (ii) is the only substantive one, and it is what an append-only instruction stream enforces: the process may not revise what it has already emitted, and what it has not written down it must remember.

Definition 2 (Dispersed family, canonical sharing, tuned modulus). With $Q, \Delta, K, m, V, P_j, U_x$ as in Sec. 2, an *r-round dispersed stream* for the aggregate $x \in [K]^m$ is any $a^{(1)}, \dots, a^{(r)} \in \mathbb{Z}_Q^m$ with $\sum_t a^{(t)} \equiv x \pmod{Q}$; correctness is required on every such stream, not only on typical ones. The *canonical sharing* $\mathcal{D}$ draws $a^{(1)}, \dots, a^{(r-1)}$ i.i.d. uniform on $\mathbb{Z}_Q^m$ and sets $a^{(r)} := x - \sum_{t < r} a^{(t)}$. The *tuned modulus* at accuracy ε is $Q_\varepsilon := \lfloor \pi / (2 \arcsin 2\varepsilon) \rfloor$.

Lemma 6 (Tuned modulus). *For $0 < \varepsilon \leq 1/8$ one has $Q_\varepsilon \geq 6$, $\sin(\pi/2Q_\varepsilon) \geq 2\varepsilon$, and $\log_2 K_\varepsilon = L - \log_2(4/\pi) + o(1) = L - 0.35 + o(1)$.*

Proof. $Q_\varepsilon \leq \pi / (2 \arcsin 2\varepsilon)$ gives $\pi/2Q_\varepsilon \geq \arcsin 2\varepsilon$, i.e. $\sin(\pi/2Q_\varepsilon) \geq 2\varepsilon$, which is the accuracy condition of Sec. 2 with a factor 2 to spare; $\varepsilon \leq 1/8$ gives $\arcsin 2\varepsilon \leq \arcsin \frac{1}{4} < \pi/12$ and so $Q_\varepsilon \geq 6$. Since $\arcsin 2\varepsilon = 2\varepsilon(1 + O(\varepsilon^2))$, $Q_\varepsilon = \pi/(4\varepsilon) + O(1)$ and $\log_2 K_\varepsilon = \log_2(Q_\varepsilon - 1) = L + \log_2(\pi/4) + o(1)$. $\square$

Lemma 7 (Projective distance). *For unitaries A, B of equal size, $d_{\text{proj}}(A, B) = 2 \sin(\Theta/4)$, where $\Theta \in [0, 2\pi]$ is the length of the shortest closed arc of the unit circle containing $\text{spec}(B^\dagger A)$.*

Proof. By unitary invariance of the operator norm, $d_{\text{proj}}(A, B) = \inf_\phi \|B^\dagger A - e^{i\phi} I\| = \inf_\phi \max_{\lambda \in \text{spec}(B^\dagger A)} |\lambda - e^{i\phi}|$, since $B^\dagger A$ is normal. Write $d(\phi, \lambda) \in [0, \pi]$ for angular distance, so that $|\lambda - e^{i\phi}| = 2 \sin(d(\phi, \lambda)/2)$, increasing in d . If $e^{i\phi_0}$ is the midpoint of a shortest arc, every eigenvalue is within $\Theta/2$ of it and the two ends attain that, so the maximum equals $2 \sin(\Theta/4)$. Conversely, for any ϕ put $D := \max_\lambda d(\phi, \lambda) \leq \pi$; the spectrum then lies in the closed arc of length $2D$ centred at $e^{i\phi}$, so $\Theta \leq 2D$ by minimality and $\max_\lambda |\lambda - e^{i\phi}| = 2 \sin(D/2) \geq 2 \sin(\Theta/4)$. $\square$

Proposition 2 (Block compiler). *For every $q \leq m$ there is a deterministic one-pass process which, on every r-round dispersed stream, keeps the running sums modulo Q of q designated coordinates in a snapshot of at most $\lceil q \log_2 Q \rceil + O(\log(mQr))$ bits and, at round r , knows the exact aggregate x_j of each designated coordinate.*

Proof. The state carried across a cut is the vector of partial sums in $\mathbb{Z}_Q^q$, encoded as one integer in $[Q^q]$, i.e. $\lceil q \log_2 Q \rceil$ bits; the remainder of the snapshot is the round index, the position within the round and a fixed-size program counter, $O(\log(mQr))$ bits, made prefix-free as in Definition 1 (iii). The designated set is fixed in advance and need not be stored. Each arriving share is added modulo Q into its partial sum, so after round r the j -th sum is $\sum_t a_j^{(t)} \pmod{Q} = x_j$. $\square$

The gap between $\log_2 Q$ here and the $\log_2 K$ of the lower bounds is the difference between storing a share, which ranges over $\mathbb{Z}_Q$, and the entropy of an aggregate, which ranges over $[K]$; it is $\log_2(Q/K) = O(1/Q)$ per stored coordinate.

B Proofs of the lemmas

Proof of Lemma 1. By Lemma 7, $d_{\text{proj}}(A, B) = 2\sin(\Theta/4)$ where $\Theta \in [0, 2\pi]$ is the length of the shortest closed arc containing $\text{spec}(B^\dagger A)$. The spectrum of $\bigotimes_j B_j^\dagger A_j$ consists of the products of eigenvalues. Fix j and two eigenvalues λ, λ' of $B_j^\dagger A_j$ at circular distance $\theta_j \leq \pi$; multiplying both by a fixed product μ of eigenvalues of the other factors gives two points of the joint spectrum at circular distance θ_j , and any closed arc containing both has length at least θ_j . Hence $\Theta \geq \theta_j$, and since $\sin$ increases on $[0, \pi/2]$, $d_{\text{proj}}(A, B) \geq 2\sin(\theta_j/4) = d_{\text{proj}}(A_j, B_j)$. For the right inequality replace one factor at a time: $\|\bigotimes_j A_j - e^{i\sum \phi_j} \bigotimes_j B_j\| \leq \sum_j \|A_j - e^{i\phi_j} B_j\|$, then take infima. For the closed form, with eigenphases a, b of $B^\dagger A$ one has $|\text{tr}|^2 = 2 + 2\cos\Theta$ and $2\sin(\Theta/4) = \sqrt{2 - 2\cos(\Theta/2)} = \sqrt{2 - |\text{tr}|}$, using $\cos(\Theta/2) \geq 0$ for $\Theta \leq \pi$. $\square$

Proof of Lemma 2. Every Clifford+ T unitary has a unique normal form $(T|\epsilon)(HT|SHT)^*C$ with C one of the 24 Cliffords modulo phase [8], and the normal form is T -optimal [9]. Strings with exactly t letters T number 2^{t-1} (initial T present) plus 2^t (absent), times 24. For tuples, $\prod_j n_1(t_j) \leq 36^m 2^{\sum_j t_j}$ and the number of (t_j) with $\sum t_j \leq T$ is $\binom{T+m}{m}$. $\square$

Proof of Lemma 3. Encode M by the Elias- γ code of T (at most $2\log_2(T+1)+1$ bits) followed by an index among $N_m^{\leq}(T)$ tuples; $H(M)$ is at most the expected length of this prefix code. Use $\log_2 \binom{T+m}{m} \leq m \log_2(e(1+T/m))$, which is concave in T , and Jensen. $\square$

Proof of Lemma 5. Write the unitary as $2^{-k/2} \begin{pmatrix} u & -t^\dagger \omega^l \\ t & u^\dagger \omega^l \end{pmatrix}$ with $u, t \in \mathbb{Z}[\omega]$, $\omega = e^{i\pi/4}$, $uu^\dagger + tt^\dagger = 2^k$ [9]. By the closed form of Lemma 1, closeness to a diagonal unitary forces $|t|^2 \leq 2\epsilon^2 2^k$. The Galois conjugate $\sqrt{2} \mapsto -\sqrt{2}$ of the norm equation gives $|t^\bullet|^2 \leq 2^k$. Hence $\xi := tt^\dagger \in \mathbb{Z}[\sqrt{2}]$ is totally non-negative with $N_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(\xi) = \xi\xi^\bullet \leq 2\epsilon^2 4^k < 1$, which forces $\xi = 0$ and $t = 0$. Then $uu^\dagger = 2^k$; the prime above 2 in $\mathbb{Z}[\omega]$ is $(1-\omega)$ with $(1-\omega)^4 \sim 2$, and the units of relative norm one are the ω^j , so $u = \omega^j(1-\omega)^{2k}$ times a unit and the unitary is $\text{diag}(\omega^a, \omega^b)$ modulo phase. $\square$

C Proof of Theorem 2

Hopf reduction. Lift W, G_1, G_2 to $\text{SU}(2)$; d_{proj} does not see the phase. Put $W' := G_1^\dagger W G_2^\dagger = \begin{pmatrix} a & -b \\ b & \bar{a} \end{pmatrix}$. Then $\text{tr}(R_z(\theta)^\dagger W') = 2\text{Re}(e^{i\theta/2}a)$, so by Lemma 1,

$$\min_\theta d_{\text{proj}}(W', R_z(\theta)) = \sqrt{2 - 2|a|} \text{ and}$$

$$\begin{aligned} \exists \theta : d_{\text{proj}}(W, G_1 R_z(\theta) G_2) &\leq \epsilon \\ \iff |b(W')|^2 &\leq \epsilon^2 (1 - \frac{\epsilon^2}{4}). \end{aligned} \quad (18)$$

Let $H_1 := G_1 S^1 G_1^{-1}$ be the maximal torus containing $G_1 R_z G_1^{-1}$ and $\pi : \text{PU}(2) \rightarrow H_1 \backslash \text{PU}(2) \cong S^2$ the Hopf map, with the measure on S^2 normalized to 1. The function $W \mapsto |b(G_1^\dagger W G_2^\dagger)|$ is left- H_1 -invariant, hence a function of $\pi(W)$, and its sublevel sets are the spherical caps centred at $\pi(G_1 G_2)$, the point through which the fibre is the coset C . Since $|b|^2$ is uniform on $[0, 1]$ under Haar measure,

$$T_\epsilon(C) \subseteq \{W : |b(G_1^\dagger W G_2^\dagger)|^2 \leq \epsilon^2\} = \pi^{-1}(\text{cap}), \quad (19)$$

a cap of measure ϵ^2 , i.e. of angular radius θ_ϵ with $\sin^2(\theta_\epsilon/2) = \epsilon^2$.

Majorant. Let $r := 1 - 2\epsilon \in [\frac{3}{4}, 1)$ and let

$$P_r(u) = \sum_{\ell \geq 0} (2\ell + 1) r^\ell P_\ell(u) = \frac{1 - r^2}{(1 - 2ru + r^2)^{3/2}} \quad (20)$$

be the Poisson kernel, a positive zonal function on S^2 of mean 1, decreasing in the polar angle. Because $1 - 2r \cos \theta_\epsilon + r^2 = (1 - r)^2 + 4r \sin^2(\theta_\epsilon/2) = 4\epsilon^2(1 + r)$ and $1 - r^2 = 2\epsilon(1 + r)$, its value at the rim of the cap is $P_r(\cos \theta_\epsilon) = 1/(4\epsilon^2 \sqrt{1 + r})$. Set

$$F_+ := a_\epsilon (P_r \circ \langle \cdot, x_0 \rangle) \circ \pi, \quad a_\epsilon := 4\epsilon^2 \sqrt{1 + r} \leq 4\sqrt{2} \epsilon^2, \quad (21)$$

with x_0 the centre of the cap. Then $F_+ \geq 0$ everywhere, $F_+ \geq 1$ on $\pi^{-1}(\text{cap})$, F_+ is left- H_1 -invariant, and $\mu(F_+) = a_\epsilon$. No smoothing or truncation is needed: the coefficients r^ℓ decay geometrically at scale $\ell \sim 1/\epsilon$, which is exactly the spectral width a cap of radius ϵ forces (Remark 2), and they are explicit.

Hecke reduction. Λ_k is closed under inversion (T -count is symmetric). Write $\Lambda_k = \bigsqcup_i \gamma_i U$, one left coset per vertex at distance $\leq k$, and let $\bar{F}(g) := \frac{1}{24} \sum_{u \in U} F_+(ug)$, which is left- U -invariant. Then

$$\begin{aligned} \sum_{\gamma \in \Lambda_k} F_+(\gamma) &= \sum_{\gamma \in \Lambda_k} F_+(\gamma^{-1}) = \sum_i \sum_{u \in U} F_+(u^{-1} \gamma_i^{-1}) \\ &= 24 \sum_i \bar{F}(\gamma_i^{-1}) = 24 \sum_{j \leq k} (T_j \bar{F})(e). \end{aligned} \quad (22)$$

Isotypic decomposition. By Peter-Weyl, $L^2(\text{SU}(2)) = \bigoplus_n V_n \otimes V_n^*$, and only even $n = 2\ell$ occur on $\text{PU}(2)$; the decomposition is preserved by left and right translations, hence by T_j and by left- U -averaging. Since F_+ is left- H_1 -invariant, its n -th component is a rank-one matrix coefficient $F_n(g) = \langle v_0, \pi_n(g) w_n \rangle$ with v_0 the unique H_1 -fixed unit vector, and it corresponds to the degree- ℓ spherical-harmonic component f_ℓ of F_+ viewed on S^2 , with $\|F_n\|_2 = \|f_\ell\|_{L^2(S^2)}$ and $n+1 = 2\ell+1$. Left translations act on the v side only, so $T_j \bar{F}_n(g) = \langle v', \pi_n(g) w_n \rangle$ is

again rank one, and rank-one coefficients satisfy $\|\phi\|_\infty \leq \sqrt{n+1} \|\phi\|_2$. For $n \geq 1$, using (R) and $\|\bar{F}_n\|_2 \leq \|F_n\|_2$,

$$|(T_j \bar{F}_n)(e)| \leq \sqrt{n+1} \|T_j \bar{F}_n\|_2 \leq \sqrt{n+1} (j+1) 2^{j/2} \|F_n\|_2. \quad (23)$$

The $n = 0$ term contributes $24 \sum_{j \leq k} (\# \text{vertices at distance } j) \mu(F_+) = N_k a_\varepsilon \leq 72 \cdot 4\sqrt{2} \cdot 2^k \varepsilon^2$, which is the main term with $C_1 = 72 \cdot 4\sqrt{2} \leq 408$.

Summation. Two geometric series finish the proof. First $\sum_{j \leq k} (j+1) 2^{j/2} \leq (1 - 2^{-1/2})^{-1} (k+1) 2^{k/2}$. Second, the degree- ℓ component of $a_\varepsilon P_r$ is $a_\varepsilon r^\ell (2\ell+1) P_\ell$, and $\|P_\ell\|_{L^2(S^2)} = (2\ell+1)^{-1/2}$, so with $(1-r)^2 = 4\varepsilon^2$,

$$\begin{aligned} \sum_{n \geq 1} \sqrt{n+1} \|F_n\|_2 &= a_\varepsilon \sum_{\ell \geq 1} (2\ell+1) r^\ell \\ &\leq a_\varepsilon \frac{1+r}{(1-r)^2} = (1+r)^{3/2} \leq 2^{3/2}. \end{aligned} \quad (24)$$

Collecting, $\#(\Lambda_k \cap T_\varepsilon(C)) \leq \sum_{\gamma \in \Lambda_k} F_+(\gamma) \leq C_1 2^k \varepsilon^2 + C_2 (k+1) 2^{k/2}$ with

$$C_2 = 24 (1 - 2^{-1/2})^{-1} 2^{3/2} = 231.7 \leq 232. \quad (25)$$

Taking $1-r = \lambda\varepsilon$ with $\lambda \neq 2$ trades the two constants against each other, $C_1 = 72(\lambda^2 + 4r)^{3/2}/(\lambda(1+r))$ and $C_2 = 24(1 - 2^{-1/2})^{-1}(\lambda^2 + 4r)^{3/2}/\lambda^3$; only $\log_2 C_2$ enters the bounds of Sec. 5, through A_2 , and it varies by one bit over $\lambda \in [2, 4]$. $\square$

D Proofs of Theorems 3 and 5

Lemma 8 (Cost-weighted mutual information). *Let (A, W) be jointly distributed with $|\text{supp } A| \leq K$; let t be a cost on the alphabet of W with $\#\{w : t(w) \leq \tau\} \leq n(\tau)$ for every τ , and let $\tau_* \geq 2 \log_2 K$ be an integer. Then*

$$I(A; W) \leq \frac{1}{2} \mathbb{E}[t(W)] + 1 + \log_2 Z, \quad (26)$$

where $Z := \sum_{\tau \leq \tau_*} n(\tau) 2^{-\tau/2}$.

Proof. Let $E := \mathbf{1}\{t(W) > \tau_*\}$, a function of W , and $q := \Pr[E = 1]$. Then $I(A; W) = I(A; W, E) \leq H(E) + (1-q)I(A; W | E = 0) + qI(A; W | E = 1)$, and $I(A; W | E = 1) \leq H(A | E = 1) \leq \log_2 K$ because A takes at most K values. On $E = 0$ the variable W is supported on $\{t \leq \tau_*\}$, so the Gibbs inequality $H(W) \leq \lambda \mathbb{E}[t] + \log_2 \sum_w 2^{-\lambda t(w)}$ with $\lambda = \frac{1}{2}$ gives

$$I(A; W | E = 0) \leq H(W | E = 0) \leq \frac{1}{2} \mathbb{E}[t | E = 0] + \log_2 Z,$$

because $\sum_{t(w) \leq \tau_*} 2^{-t(w)/2} \leq \sum_{\tau \leq \tau_*} n(\tau) 2^{-\tau/2} = Z$. Finally $\mathbb{E}[t] \geq (1-q)\mathbb{E}[t | E = 0] + q\tau_*$ and $\log_2 K \leq \tau_*/2$, so $(1-q)\frac{1}{2}\mathbb{E}[t | E = 0] + q\log_2 K \leq \frac{1}{2}\mathbb{E}[t]$, while $H(E) \leq 1$. $\square$

The choice $\tau_* = \lceil 2 \log_2 K \rceil$ is the smallest admissible one and the best: Z grows with τ_* . With the profile of Corollary 2,

$$Z \leq \frac{C_1 \varepsilon^2 2^{\tau_*/2}}{1 - 2^{-1/2}} + \frac{C_2 (\tau_* + 1) (\tau_* + 2)}{2}, \quad (27)$$

in which the first term is $O(\varepsilon)$ for the tuned family; at $\varepsilon = 10^{-10}$, $\tau_* = 66$ and $A_2 = 1 + \log_2 Z = 20.01$.

Lemma 9 (Cost-weighted entropy). *Let W take values in an alphabet with cost $t : \mathcal{A} \rightarrow \mathbb{N}$ such that $\#\{W : t(W) \leq \tau\} \leq c_0 + c_1 \tau + c_2 \tau^\beta$ for all τ . Put $\tau_0 := \lfloor \beta L - \log_2 c \rfloor$, $\mathcal{A}_0 := \{t \leq \tau_0\}$, $N_0 := c_0 + c_1 \tau_0 + 1$ (so $|\mathcal{A}_0| \leq N_0$) and $q := \Pr[W \notin \mathcal{A}_0]$. Then*

$$\begin{aligned} H(W) &\leq \mathbb{E}[t(W)] - q(\beta L - \log_2 c - 1) \\ &\quad + h_2(q) + \log_2 N_0 + 3 \log_2(1 + \mathbb{E}t). \end{aligned} \quad (28)$$

Proof. $H(W) \leq h_2(q) + (1-q) \log_2 N_0 + qH(W | W \notin \mathcal{A}_0)$. On the complement code $t - \tau_0$ in Elias- γ and an index among the words of cost exactly t , of which there are at most $c_0 + c_1 t + c_2 t^\beta \leq (c_0 + c_1 t + 1) c_2 t^\beta$ (for $t > \tau_0$, $c_2 t^\beta \geq 1$), and $\log_2(c_0 + c_1 t + 1) \leq \log_2 N_0 + \log_2(1 + t - \tau_0)$ because $c_1 \leq N_0/\tau_0$: length $\leq t - \beta L + \log_2 c + \log_2 N_0 + 3 \log_2(t - \tau_0 + 1) + 1$. Then $q\mathbb{E}[t | \notin] \leq \mathbb{E}t$ and $q \log_2(1 + a) \leq \log_2(1 + qa)$. $\square$

Proof of Theorem 3. As in Theorem 1, A is uniform on a product translate given Z , so the A_j are conditionally independent given Z . Per-coordinate Fano (the decoded $\hat{X}_j$ is a function of $(\bar{W}_j^{(t)}, \sigma_t, Z)$ and errs with probability $\leq \delta$): $I(A_j; \bar{W}_j, \sigma_t | Z) \geq \ell_\delta$. By the chain rule and conditional independence,

$$\begin{aligned} m\ell_\delta &\leq \sum_j I(A_j; \sigma_t | Z) + \sum_j I(A_j; \bar{W}_j | \sigma_t, Z) \\ &\leq S + \sum_j I(A_j; \bar{W}_j | \sigma_t, Z), \end{aligned} \quad (29)$$

where $\sum_j I(A_j; \sigma_t | Z) \leq I(A; \sigma_t | Z) \leq H(\sigma_t) \leq S$ because $I(A_j; \sigma | Z, A_{<j}) \geq I(A_j; \sigma | Z)$ under conditional independence.

Fix j and (s, z) , write $t := t_j^{(t)}$, and let E_j be the indicator that coordinate j is correct, $\delta_j := \Pr[E_j = 0 | s, z]$. Then

$$\begin{aligned} I(A_j; \bar{W}_j | s, z) &\leq h_2(\delta_j) + (1 - \delta_j) I(A_j; \bar{W}_j | E_j = 1, s, z) + \delta_j \log_2 K. \end{aligned} \quad (30)$$

On $E_j = 1$ the alphabet of $\bar{W}_j$ obeys the profile of Corollary 2, and A_j takes at most K values, so Lemma 8 with $\tau_* = \lceil 2 \log_2 K \rceil$ gives

$$I(A_j; \bar{W}_j | E_j = 1, s, z) \leq \frac{1}{2} \mathbb{E}[t | E_j = 1, s, z] + A_2. \quad (31)$$

Using $(1 - \delta_j)\mathbb{E}[t \mid E_j = 1, s, z] \leq \mathbb{E}[t \mid s, z]$,

$$I(A_j; \bar{W}_j \mid s, z) \leq \frac{1}{2}\mathbb{E}[t_j \mid s, z] + A_2 + h_2(\delta_j) + \delta_j \log_2 K. \quad (32)$$

Average over (s, z) (Jensen for h_2), sum over j , and use $\sum_j \bar{\delta}_j \leq m\delta$: $m\ell_\delta \leq S + \frac{1}{2}\bar{T}_t + mA_2 + mh_2(\delta) + \delta m \log_2 K$. Rearranging with $\ell_\delta = (1 - \delta) \log_2 K - h_2(\delta)$ gives the claim. $\square$

Proof of Theorem 5. Start from (29). Fix a coordinate j and a context (s, z) ; let E_j be the event that coordinate j is correct, $d_j := \Pr[E_j = 0 \mid s, z]$, and $q_j := \Pr[t_{\min}(\bar{W}_j) > \tau_0 \mid E_j = 1, s, z]$. Write $k := \log_2 K$ and $g_\delta := h_2(\delta) + \delta k$. On $E_j = 1$ the word lies, by Conjecture 1 applied to the coset of Corollary 2, in an alphabet whose cost- $\leq \tau_0$ part has at most N_0 elements. Splitting first on E_j and then on the high-cost event,

$$I(A_j; \bar{W}_j \mid s, z) \leq h_2(d_j) + d_j k + (1 - d_j)[1 + \log_2 N_0 + q_j k].$$

Averaging over contexts, summing over j and inserting into (29) gives $m\ell_\delta - S \leq mg_\delta + ma_0 + k\bar{\mathcal{A}}_t^\geq$ with $\bar{\mathcal{A}}_t^\geq = \sum_j (1 - d_j)q_j$, which is (14) because $m\ell_\delta - mg_\delta - S = D_t$. For the second bound apply Lemma 9 to the alphabet conditional on $E_j = 1$:

$$I(A_j; \bar{W}_j \mid E_j = 1, s, z) \leq \mathbb{E}[t \mid E_j = 1, s, z] - bq_j + 1 + \log_2 N_0 + 3 \log_2 (1 + \mathbb{E}[t \mid E_j = 1, s, z]).$$

Multiply by $1 - d_j$, average over contexts and sum over j : $(1 - d_j)\mathbb{E}[t \mid E_j = 1, s, z] \leq \mathbb{E}[t \mid s, z]$, the concavity of $x \mapsto \log_2(1 + x)$ bounds the logarithmic terms by $3m \log_2(1 + \bar{T}_t/m)$, and the high-cost terms sum to $b\bar{\mathcal{A}}_t^\geq$, so $D_t \leq \bar{T}_t - b\bar{\mathcal{A}}_t^\geq + ma_0 + 3m \log_2(1 + \bar{T}_t/m)$. Since $b \geq 0$, substituting (14) yields (13). Summing over $t < r$ and using concavity of the logarithm gives the statement for T_{pre} ; at $\delta = 0$ and $S = 0$ the additive terms are $O(m \log L)$ per round. $\square$

E Proof of Theorem 7

Let Ω be the success event; on Ω , u is a function of the message (snapshot and unitary tuple) together with (v, y) , and the message takes at most $2^{S_{\max}} N_m^{\leq}(T_{\max})$ values; Lemma 2 gives the first inequality. For the second, given Ω let $P' = P \cdot 1_\Omega$, a sub-normalized distribution within δ of P ; then $H_{\max}(U \mid VY)_{P'} \leq \log_2 \max_{v,y} |\text{supp } P'_{U|v,y}|$, so $H_{\max}^\delta \leq H_0^\delta$. $\square$

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